Reliable and Reproducible Demographic Inference for Fairness in Face Analysis

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Abstract

Fairness evaluation in face analysis systems (FAS) uses automatic demographic attribute inference (DAI), which in turn relies on predefined demographic segmentation. However, the validity of fairness auditing hinges on the reliability of the DAI process, a dependency we prove theoretically – and assert empirically – by showing that improved DAI reliability leads to less biased, lower-variance estimates of FAS fairness. We revisit DAI by proposing REDI-Face, a fully reproducible DAI pipeline that replaces conventional end-to-end training with a modular transfer learning approach. Our design integrates pretrained face recognition encoders with non-linear classification heads. We audit this pipeline across three dimensions: accuracy, fairness, and a newly introduced notion of robustness, defined via intra-identity consistency. The proposed robustness metrics apply to any demographic segmentation scheme. We benchmark our pipeline REDI-Face on gender and ethnicity inference across multiple datasets and training setups, and show that it outperforms strong baselines, particularly on ethnicity, which is the more challenging attribute. The accuracy improvement is between 7 and 10 points, while fairness and robustness are at least doubled compared with FairFace, the best baseline. To promote reproducibility and adoption, we will release the training dataset metadata, full codebase, pretrained models, and evaluation toolkit.

1. Introduction

The accuracy of automatic face analysis systems (FAS) has significantly progressed over the past decade, driven by increasingly sophisticated deep learning architectures [11, 28] and the availability of large-scale training datasets [68]. FAS now powers critical applications, from smartphone authentication to national security infrastructure [6, 56, 61]. Given this societal impact, it must be not only accurate but also fair to ensure responsible deployment and public trust [46]. Numerous studies have shown that FAS performance can vary significantly across demographic groups [7, 15, 19, 20, 30, 51]. These disparities are often attributable to demographic imbalance in training data [20, 64], limited data diversity [15], visual variability [59], or suboptimal training procedures [31]. While these studies have advanced our understanding of algorithmic bias, they largely overlook a foundational assumption: the reliability of the demographic attribute inference (DAI) process itself.

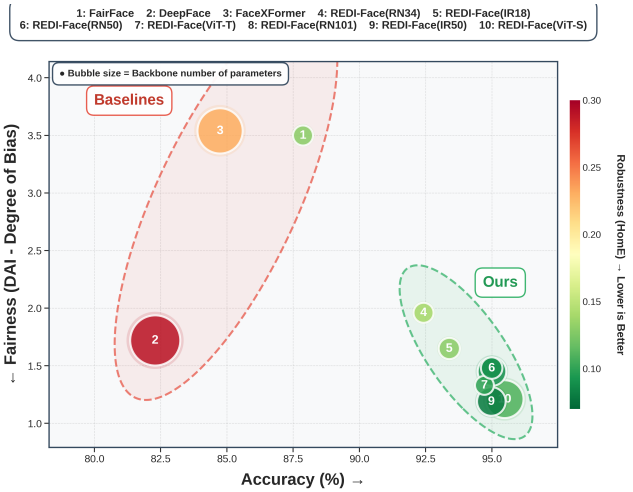


Figure 1. Ethnicity inference results (4 classes) audited using accuracy, fairness, and robustness dimensions on the BFW dataset [49] for baselines (FairFace [26], DeepFace [54] and FaceXFormer [40]) and several instances of the proposed approach. Higher values are better for accuracy. **Lower** values are better for **fairness**. **Greener** is better for robustness. The circle sizes are proportional to the number of backbone DNN parameters, with smaller circles indicating improved inference efficiency. An ideal DAI system would be represented by a collapsed circle located at the bottom-right corner of the plot.

In practice, attribute inference typically relies on off-the-shelf models [30], with FairFace [26] being the widely used. Yet, as shown in Figure 1, FairFace offers suboptimal performance and compromises the fairness evaluation of downstream systems. Compounding the issue, FairFace is difficult to audit: its training code is unavailable, and several reproduction attempts have failed [1].

[1] https://github.com/joojs/fairface/issues

Our first contribution is a theoretical justification for the need for reliable DAI. Surprisingly, to our knowledge, no formal link between DAI reliability and the FAS fairness validity has been established, yet we show that it is crucial. We assert empirically the theoretical links established.

Our second contribution is a fully transparent and reproducible pipeline that audits DAI performance across several dimensions. Transparency mainly implies a use of publicly available resources, that was not always respected in previous approaches. Reproducibility requires but should not rely on providing the code only, with possibly many hyperparameters not studied in the paper. We argue that a simple model design favors reproducibility. For this reason, we design REDI-Face, a modular design: we pretrain or reuse face recognition backbones and train only the classification heads with an RBF kernel. This kernel has the advantages of modelling through a non-stochastic training, the non-linearity within the considered latent space, and minimizing the number of hyperparameters. Using publicly available data, we conduct extensive experiments for ethnicity and gender inference, varying backbones, dataset configurations, and classification heads. Remarkably, this intentionally simple approach not only guarantees reproducible results but also outperforms several existing state-of-the-art methods: the ethnicity inference results in Figure 1 show significant accuracy, fairness, and robustness improvements for our approach compared with FairFace [26], DeepFace [54] and FaceXFormer [40] or even smaller structures, thanks to the well-adapted pretraining.

Finally, we advocate for conducting DAI performance audits across three dimensions: accuracy, fairness, and robustness. Optimizing all three DAI dimensions is essential for valid FAS fairness auditing: accuracy underpins correct interpretations; fairness ensures equitable error distribution across groups; robustness, an underexplored property, promotes intra-identity consistency. To estimate this last dimension, we introduce novel metrics based on the simple hypothesis: *Given N images of the same identity, an ideal demographic classifier should predict the same segment for all samples.* In particular, we formalize it through an entropy-based robustness metric that generalizes across demographic taxonomies.

This work advances demographic attribute inference by making it more reliable and reproducible - two qualities that are critical for valid fairness evaluations in face analysis. Our simple yet high-performing pipeline questions current DAI practices. We hope it will contribute as a reliable foundation for future developments that will further improve DAI accuracy, fairness, and robustness.

2. Related Work

Fairness evaluation [7, 15, 19, 20, 45, 51] is needed to comprehensively audit AI-driven face analysis systems deployed in practical applications, including border control [23], consumer authentication [58], public safety [38], face-based payment [48]. These studies highlight significant performance variation across demographic segments, due to imbalance in face recognition training datasets [20, 64], insufficient data diversity [15], or biases in the architectures or training strategies [13, 39, 47]. While understanding and mitigating biases is important, little effort is devoted to the reliability of the demographic attribute inference that enables their conclusions. Below, we analyze the main components of existing inference pipelines, as well as other relevant works.

The demographic segmentation scheme represents the basis for any demographic classifier. The dominant approach [7, 26, 62, 63] defines a small set of predefined ethnicity categories. Past works proposed alternatives to the ethnicity-based categorization used in FairFace, notably based on skin color [9]. The DiF dataset [37] implemented a facial coding scheme consisting of skin color and six other attributes describing the face shape. More recently [34, 60] modeled skin appearance by adding the hue angle to the tone measured using the Fitzpatrick scale. These segmentations aim to encode medical and/or anthropological attributes in face coding schemes, rather than predefining ethnic categories. This modeling choice makes them appealing alternatives, but existing studies do not test their robustness. In addition, their semantic interpretation is less direct compared to that of predefined categories. We note that all demographic segmentation schemes are criticizable because they categorize the vast diversity of human faces into a limited set of categories. The sensitive nature of face data amplifies the difficulty [30]. However, attribute segmentation is necessary to facilitate downstream evaluation. We argue that we should select a particular scheme based on a comprehensive analysis of its accuracy, robustness, and fairness.

Training datasets are a core component of demographic classification. In supervised settings, their annotation follows a predefined segmentation. Representative examples include the RFW [62], BFW [49], FairFace [26], BUPT [63]. Their common characteristics include the manual curation of demographic annotations and the emphasis on segment balancing. While balancing usually increases fairness, multiple studies [3, 15, 31, 51] show that it only partially addresses DAI unfairness. The authors of [59] investigate the contribution of non-demographic factors on face recognition biases. Different works adapted the training process to improve fairness: SensitiveLoss [55], a discrimination-aware loss, penalizes demographic disparities, GAC [18] adapts convolution kernels and attention mechanisms to faces based on their demographic attributes, and DebFace [17] uses adversarial learning to reduce face recognition and DAI biases jointly, while [4] tackles attribute combinations during training. Interestingly, despite

strong efforts devoted to face generation [29, 34], fairness issues subsist and are even amplified compared with real data [15]. The authors of [64] examine the dataset structure (number of identities and of images per identity), but also factors DAI fairness, such as head pose, brightness, and image quality. These studies inform our experiments, and we test different training dataset structures and sizes.

FairFace [26], the work closest to ours, is a widely used demographic classification tool [10, 12, 15, 32, 36, 51] that jointly infers ethnicity, gender, and age. The authors demonstrate that training on their balanced dataset yields more equitable performance across demographic groups compared to models trained on imbalanced alternatives. One important limitation is that the method achieves 88% 4-ethnicity classification accuracy on the RFW dataset, which affects the relevance of fairness auditing based on it. Beyond performance limitations, FairFace suffers from a critical reproducibility gap. The authors have not released training code or sufficient implementation details to enable independent replication, despite numerous attempts [2], including ours. This opacity undermines the scientific reproducibility that is essential for bias auditing, leaving practitioners unable to verify claimed performance or adapt the methodology to new contexts. More recently, the DeepFace [54] framework includes DAI capabilities as well as FaceXFormer [40], trained on a merge of the FairFace [26] and UTKFace [66] datasets to improve global accuracy. We use FairFace, DeepFace, and FaceXFormer as baselines in our experiments.

3. Demographic attribute inference

We first motivate the need for reliable DAI pipelines by theoretically analyzing the effect of accuracy in downstream face analysis tasks whose fairness needs evaluation. This analysis is motivated by the observation that, for a given model, most existing fairness metrics do not correlate with DAI accuracy. For instance, if applied to a sufficiently large and balanced test dataset, a random classifier obtains a nearly perfect degree of bias scores since the accuracy of all demographic segments is almost identical. However, we demonstrate that the DAI accuracy has significant implications for estimating the fairness model in addressing a downstream task. Then, we present the main components of the DAI pipeline, summarized in Figure 2, namely the training datasets, DAI model training approaches, and evaluation dimensions.

3.1. Theoretical Motivation

We consider a face analysis system that performs face recognition (FR), face verification (FV), or other similar tasks. It is trained on an image dataset annotated with task-dependent labels, e.g., a class number associated with each image for recognition or a binary value for FV image pairs. These annotations are used to estimate the performance of the face analysis system [45], with metrics such as micro-average accuracy, True Match Rate (TMR), or False Match Rate (FMR) (see Appendix B in the appendix for details).

Importantly, each image also has sensitive attribute annotations, such as the age, gender, or ethnicity of the person represented. These attributes are required to estimate the fairness with metrics such as the Degree of Bias (DoB) [16], the Demographic Parity Difference (DPD) and Demographic Parity Ratio (DPR) [1, 2], or the Equalized Odds Difference (EOD) and Equalized Odds Ratio (EOR) [1, 21]. Interestingly, most, if not all, of these fairness metrics can be expressed from a binary indicator Y_i, allowing us to derive a theoretical framework which is valid regardless of the metric further used (see details in subsection C.4 in the appendix).

In large-scale face datasets, annotating such attributes is tedious; thus, one often relies on automatic attribute classifiers to estimate them. These annotations are therefore prone to errors, which may further influence the quality of the fairness evaluation. To our knowledge, existing works do not address this point. In the following, we establish a plug-in estimator of the performance indicator conditioned on the *estimated* sensitive attribute. We establish an upper bound on its bias relative to the true sensitive attribute as well as the expression of the covariance matrix of the unbiased estimator. Globally, this clarifies how a more accurate estimation of sensitive attributes enhances the fairness estimation of the face analysis system.

Let $G \in \{1, \dots, K\}$ be the true latent group of an identity, $\widehat{G} \in \{1, \dots, K\}$ the estimated label of the corresponding sensitive attribute, and $Y \in \{0, 1\}$ a binary performance indicator associated to an identity for a given automatic FAS. For instance, G could be the group `old`$\times$ `caucasian` $\times$ `male` or `young` $\times$ `indian` $\times$ `female`. Y is a generic FAS performance indicator that can be specialized to reflect various metrics such as the False Match Rate (see appendix subsection C.4).

We define the prior of the sensitive attribute a as $\pi_a := \Pr(G = a)$ and the probability of success of the FAS given the *true* said sensitive attribute as $p_a := \Pr(Y = 1 \mid G = a)$. This last is actually estimated automatically with some attribute classifiers that have a confusion matrix $C = (c_{ab})_{a,b=1}^{K}$ with $c_{ab} := \Pr(\widehat{G} = b \mid G = a)$ s.t each row of C sums to 1. We then define $\tau_b := \Pr(\widehat{G} = b) = \sum_{a=1}^{K} \pi_a c_{ab}$ and denote the vectors $p = (p_1, \dots, p_K)^\top$, $\pi = (\pi_1, \dots, \pi_K)^\top$, and $\tau = (\tau_1, \dots, \tau_K)^\top = C^\top \pi$. We assume that, conditional on the true group G, the estimated label $\widehat{G}$ gives no further information about Y, that is:

$$(Y \perp\!\!\!\perp \widehat{G}) \mid G \tag{1}$$

[2] https://github.com/joojs/fairface/issues

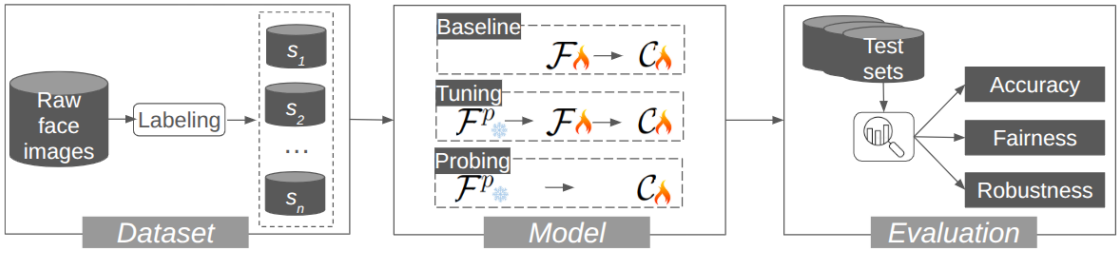


Figure 2. Overview of pipelines designed for training and evaluating demographic attribute inference. The raw data labels correspond to a demographic split schema comprising n segments. The model training component implements different strategies. We compare: (1) the baseline learning strategy implemented in FairFace[26], which jointly learns the feature extractor $\mathcal{F}$ and classifier $\mathcal{C}$, (2) the learning strategy implemented in DeepFace [54], which fine-tunes the entire pretrained FR model, and (3) training only the classifier $\mathcal{C}$ on top of the frozen $\mathcal{F}^p$. Finally, we follow common practice to evaluate the accuracy and fairness of the obtained models, and add a robustness evaluation employing a newly introduced metric.

This denotes the conditional independence of $\widehat{G}$ and Y given G. Said another way, we assume that the performance of the FAS depends on the identity and attribute group G but not on the internal randomness of the attribute classifier that predicts $\widehat{G}$ instead of G. We also assume $C^\top$ is invertible (so C is invertible). We consider an i.i.d. sample of I identities; for identity i we observe $(\widehat{G}_i, Y_i)$ but not G_i. For each observed label g we define the empirical plug-in estimator

$$\widehat{m}_g = \frac{\frac{1}{I}\sum_{i=1}^{I} \mathbf{1}\{\widehat{G}_i = g\} Y_i}{\frac{1}{I}\sum_{i=1}^{I} \mathbf{1}\{\widehat{G}_i = g\}} \tag{2}$$

We note $\widehat{m} = (\widehat{m}_1, \ldots, \widehat{m}_K)^\top$ assuming the denominator is nonzero for the K groups considered, and $\widehat{\tau}$ the empirical estimator of τ. Hence:

Proposition 1. *(informal) Under the assumptions above, the empirical plug-in estimator $\widehat{m}_g$ converges in probability toward $m_g := \Pr(Y = 1 \mid \widehat{G} = g)$ as $I \to \infty$ and $m_g = \frac{1}{\tau_g}\sum_{a=1}^{K} \pi_a c_{ag} p_a$. Moreover, the bias of the plug-in limit relative to the true group rate p_g satisfies*

$$\mathrm{Bias}_g := m_g - p_g = \frac{\sum_{a \neq g} \pi_a c_{ag} (p_a - p_g)}{\sum_{a=1}^{K} \pi_a c_{ag}}, \tag{3}$$

and therefore if $c_{ag} \to 0$ for every $a \neq g$ and $c_{gg} \to 1$ then $\mathrm{Bias}_g \to 0$. Moreover, for $\Delta_{\max,g} := \max_{a \neq g} |p_a - p_g|$,

$$\left|\mathrm{Bias}_g\right| \leq \Delta_{\max,g} \cdot \frac{\sum_{a \neq g} \pi_a c_{ag}}{\pi_g c_{gg}}. \tag{4}$$

Hence, if the confusion matrix tends towards identity – that is, the estimation of the attributes is better – the bias tends towards zero. Regarding the variance, we have:

Proposition 2. *(informal) Assuming C is invertible, the estimator*

$$\widehat{p}^{(\mathrm{corr})} = \frac{(C^\top)^{-1}(\widehat{\tau} \odot \widehat{m})}{\pi} \tag{5}$$

is asymptotically unbiased for p. Moreover the multiplicative increase in variance is controled by the factor $\|(C^\top)^{-1}\|_{\mathrm{op}}^2$, where $\|\cdot\|_{\mathrm{op}}$ is the operator norm, such that when $C \to I$ the factor tends to 1, while ill-conditioned C (small singular values) amplify the variance.

The formal propositions and their proofs are in Appendix C. The combination of both propositions shows that a better attribute inference leads to a less biased estimation of the metrics used to estimate the fairness of the FAS, with a smaller variance. This is experimentally demonstrated in Figure 3, where our pipeline, achieving the highest DAI accuracy among baselines, produces the most accurate FAS fairness metric approximations. Detailed empirical validation is provided in Appendix

3.2. Training datasets

Figure 2 provides a generic view of the DAI pipeline. The curated dataset $\mathcal{D}$ yields a set $S = s_1, s_2, ..., s_n$ of demographic segments for each attribute. For instance, ethnicity may be defined using predefined categories such as *Asian, Black, Indian, White* [26, 63], country of origin [33, 44], or skin appearance [37, 60]. Any segmentation scheme introduces ethical and technical challenges [20, 27, 44], as it reduces the diversity of human faces into a finite set of n classes. In practice, we argue that segmentation should be selected to optimize for classification accuracy, fairness, and robustness jointly. Beyond the segmentation choice itself, the dataset should contain diverse and sufficiently large samples for each segment, with reliable and high-quality labels [63].

3.3. Model training

Surprisingly, many widely used DAI models ($\mathcal{M}$), comprising a feature extractor $\mathcal{F}$ and a classifier $\mathcal{C}$, still rely on training from scratch [26] or full fine-tuning [54] of pretrained models. This persists despite the proven benefits of transfer learning in related face analysis tasks [5, 15, 41, 57].

Motivated by these findings, we explore the use of pretrained models $\mathcal{F}^p$ as an alternative to full end-to-end training. In the probing setting (Figure 2), we train the classifier $\mathcal{C}$ directly on top of a frozen backbone $\mathcal{F}^p$. Prior DAI works [24, 25] that employed pretraining typically started from ImageNet-pretrained models. However, due to the domain gap between general image classification and facial representations, this approach is suboptimal. Instead, we use $\mathcal{F}^p$ pretrained for face recognition to leverage domain-relevant representations. A common assumption [26, 51] is that the demographic segments S are linearly separable in the embedding space, motivating the use of a linear classifier $\mathcal{C}$. We challenge this assumption and instead adopt non-linear classification heads, which we show to improve performance, and formally fit with the angular loss induced structure, in Appendix.

3.4. Evaluation dimensions

DAI evaluation typically involves accuracy metrics, often complemented by fairness indicators [51]. These allow per-segment behavior analysis and deeper performance diagnostics [15]. A key challenge is to balance accuracy and fairness, as improvements in one may degrade the other [65]. We include both in our evaluation and introduce robustness as a complementary dimension.

We posit that a reliable attribute classifier should produce consistent predictions for different images of the same identity. In practice, this consistency is not guaranteed due to variations in illumination, pose, resolution, and other capture conditions [37, 60], as illustrated in Figure 1.

We define a robustness metric below, which is **taxonomy agnostic**, since it leverages not Ethnicity but Identity label. Let us consider an image dataset $\mathcal{I} = \{I_j^{id}; id \in [\![1, D]\!], j \in [\![1, N_{id}]\!]\}$, with D identities, an identitity id being represented by N_{id} images. We consider a labeling function $f_{\mathcal{A}}^{\mathcal{C}}$ that attributes a label to each image, each label being an element of a finite set $\mathcal{C}$, and depends on a dataset $\mathcal{A}$ ($f_{\mathcal{A}}^{\mathcal{C}}$ is defined below).

For each identity id, we compute the relative frequency of each possible label $k \in \mathcal{C}$ noted as $p_{id}^k = \frac{1}{N_{id}} \sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})}$. We then compute the **homogeneity entropy** as the average entropy of p over all identities:

$$HomE = \frac{1}{D} \sum_{id} \frac{1}{\log(C)} \sum_{k} -p_{id}^k \log(p_{id}^k) \qquad (6)$$

This entropy-based measure captures intra-identity consistency. Lower values of `HomE` indicate higher robustness and are thus desirable for trustworthy DAI pipelines.

4. Experiments

We first present the experimental setup used to audit demographic attribute inference. Next, we report and analyze the main results obtained with both the baseline and the proposed DAI pipelines. Finally, we present ablation experiments to study the contribution of the main components of our approach.

4.1. Experimental setup

DAI pipelines. We compare our probing-based demographic attribute inference approach against three popular baselines: FairFace [26], DeepFace [54], and FaceXFormer [40]. FairFace uses a ResNet34 (RN34) [22] architecture and combines a publicly released subset with proprietary data that is not available [26]. We also train a DAI pipeline with the FairFace public subset and the ResNet34 architecture in an attempt to reproduce the original results. The FairFace baselines implement the full training pipeline illustrated in Fig. 2. DeepFace [54] performs full fine-tuning on top of a VGGFace model [42] pretrained for face recognition. It combines training data from multiple sources and follows the fine-tuning procedure shown in Fig. 2. Similarly, FaceXFormer [40] combines multiple training sources (UTKFace, MORPH-II , and FairFace). Its particularity lies in the use of a Swin-B backbone trained on a multitask loss, whose training code hasn't been released.

Our probing-based DAI pipelines use a variety of pretrained face recognition (FR) models combined with an RBF classifier. We evaluate both CNN and transformer architectures with a comparable parametric footprint to the baselines (see Tab. 4 in supp.mat.). Specifically, we test ResNet34, ResNet50, iResNet18 (IR18) [11], and ViT-T [14], whose parameter counts range from 19.1M to 24.6M. To explore the effect of increasing model size, we additionally use ResNet101 (RN101), iResNet50 (IR50), and ViT-S [14]. We use the InsightFace implementations for transformer-based models[3]. The pretrained models are trained on a subset of the WebFace260M dataset [68], comprising approximately 28k identities and 1 million images. We selected this subset size based on GPU constraints. To assess the impact of pretraining data size, we also experiment with existing AdaFace [28] models trained on the full WebFace dataset.

We implement two types of classification heads: (1) an RBF-kernel SVM (results reported in Tab. 1) for non-linear probing, and (2) a linear SVM (reported in Appendix) for linear probing. Both classifiers are trained on a balanced subset of 100k images from BUPT-Balanced, covering 25.3k identities and balanced across gender and ethnicity. This image count matches that of the FairFace public subset [26] for fair comparison. Ethnicity labels are directly available in BUPT. For gender, we leverage the presence of multiple samples per identity and use majority voting over predictions made by FairFace. This strategy, evaluated on a 500-identity subset, yields gender label accuracy exceed-

[3] https://github.com/deepinsight/insightface

ing 99.5%, making it reliable for training and evaluation. We reserve a separate BUPT validation subset containing 60k images and 1,330 identities. All reported results use the DAI pipeline versions that maximize accuracy on this validation set. Experimentally the RBF kernel yields better results, as explained in Appendix

Test datasets. We evaluate the reliability of ethnicity and gender inference using publicly available datasets selected based on the availability of demographic attributes and identity labels. BFW [50], RFW [62], and BUPT [63] all employ a common four-class ethnicity segmentation and are used for accuracy and fairness evaluation. We sample 63k images from BUPT (1,400 identities) balanced for ethnicity. As BUPT is also used for DAI training and validation, we ensure that this test subset is completely disjoint from the training and validation sets (disjoint set of Identities).

BFW and BUPT include multiple samples per identity and are used for evaluating ethnicity robustness. BFW includes gender annotations, enabling evaluation across all three dimensions. BUPT gender labels are inferred using the same identity-based majority voting procedure applied during classifier training, ensuring consistency.

Metrics. We report micro-average for accuracy evaluation. For fairness, we use the degree of bias (DoB), computed as the standard deviation of per-group accuracies (lower is better). For robustness, we use the Homogeneity Entropy (HomE) metric defined in Eq. (6), along with additional robustness measures described in Sec. B.

4.2. Main results

Tab. 1 presents the accuracy, fairness, and robustness scores for both ethnicity and gender inference pipelines. Overall, the proposed approach achieves substantial improvements across all evaluation dimensions and datasets. Similarly to the community attempts, our own attempts to reproduce the training of FairFace yields significantly poorer results.

Ethnicity. The accuracy gains of REDI-Face (from 7 to 10 percentage points) are consistant across all architecture, even the lightest ones. Compared with FairFace [26], the degree of bias (DoB) is reduced by a factor of 2–3, and HomE is approximately halved. The comparison with DeepFace [54] is even more favorable in terms of accuracy and robustness, although DeepFace exhibits better fairness scores than FairFace despite lower overall accuracy. Although better than DeepFace for accuracy and robustness, FaceXFormer falls behind FairFace in all audit dimensions. Interestingly, performance differences between compact and larger models remain modest when using a pretrained FR model with an RBF classifier—highlighting the robustness and efficiency of this simple yet effective DAI pipeline. Among compact architectures, RN34 yields the weakest results, while RN50 and ViT-T perform comparably. For larger backbones (highlighted in blue), IR50 and ViT-S achieve slightly better accuracy and show notable improvements in fairness and robustness. Pretraining with a larger dataset (orange cells) does not consistently improve accuracy or fairness and tends to reduce robustness. The main difference lies in dataset structure: the 1M-image WebFace subset is balanced, whereas the 40M-image version is highly imbalanced. These results indicate a trade-off between dataset size and class balance, which we further investigate in Sec. 4.3. In practice, we recommend RN50 or ViT-T for computationally constrained environments and IR50 for high-performance settings.

Gender. This inference is simpler than ethnicity inference, as previously noted [26] and confirmed by results in Tab. 1. REDI-Face shows consistently strong performance, confirming the stability of the transfer-learning-based approach. Although performance differences are smaller due to task saturation, IR50—our best-performing configuration—still improves accuracy by 1.5 points on BFW and 0.5 points on BUPT compared to FairFace. In terms of fairness, our method improves performance on BFW, but FaceXFormer remains fairer on BUPT. Robustness improves across both datasets relative to FairFace. Among compact models, RN50 performs best, with ViT-T as a close second. For larger backbones, IR50 slightly outperforms ViT-S, although the differences are minor. As with ethnicity, larger models provide modest gains over compact ones. Pretraining with the larger WebFace dataset slightly improves accuracy but produces mixed effects on fairness and robustness.

Considering both ethnicity and gender performance, RN50 and IR50 emerge as the most reliable DAI pipelines among compact and large architectures, respectively. In practice, the choice between them depends on computational budget and deployment constraints.

4.3. Ablation and analysis

We perform two ablations for ethnicity inference to highlight the impact of imbalance in the pretraining dataset and that of size and structure in the BUPT subset used to train the ethnicity classifier. We then present the detailed ethnicity-wise robustness results on BFW, allowing a comparison between different taxonomies.

Pretraining dataset balancing. WebFace overrepresents White individuals, while the pretraining subset used in Sec. 4.2 is balanced. We sample 1M WebFace images randomly to assess the impact of imbalance during pretraining. Tab. 2 presents the accuracy obtained with the IR50 model. The results indicate that balancing during pretraining generally provides a slight advantage when inferring ethnicity.

Ethnicity classification dataset size and structure. Tab. 3 details the results obtained with different sizes and

DAI Pipeline	Arch.	#labels (K)	Ethnicity								Gender					
			Accuracy ↑			DAI Fairness ↓			Robustness ↓		Accuracy ↑		DAI Fairness ↓		Robustness ↓	
			BFW	RFW	BUPT	BFW	RFW	BUPT	BFW	BUPT	BFW	BUPT	BFW	BUPT	BFW	BUPT
FairFace [26]	RN34	89	87.87	88.08	82.27	3.50	4.10	4.58	0.19	0.22	96.27	98.05	2.01	*1.31*	0.13	0.09
FairFace$_R$	RN34	89	64.04	67.53	59.55	9.83	9.76	10.62	0.38	0.41	92.27	95.21	2.29	4.17	0.25	0.19
FairFace$_B$	RN34	89	78.22	77.83	89.41	4.95	6.89	2.28	0.35	0.18	93.80	96.69	2.43	4.20	0.19	0.12
DeepFace[54]	VGG-Face	?	82.29	82.31	74.37	1.72	3.15	3.03	0.29	0.36	85.43	91.22	26.88	25.55	0.31	0.16
FaceXFormer [40]	Swin-B	≈ 150	84.74	83.37	76.33	3.54	3.11	6.17	0.23	0.26	95.08	96.87	0.97	**1.14**	**0.12**	0.09
CLIP	CLIP Base		83.31	83.25	75.19	5.23	3.37	7.11	0.21	0.27	73.3	87.05	53.16	38.67	0.44	0.21
CLIP	CLIP Large		87.57	89.22	83.27	2.83	1.60	4.07	0.20	0.21	67.95	78.90	63.87	53.47	0.38	0.17
QWEN3	4B Instruct		93.62	92.80	89.05	2.28	2.40	2.87	0.11	.14	98.19	98.61	0.87	1.47	0.07	0.05
QWEN3	8B Instruct		89.71	90.94	86.14	4.82	3.81	4.26	0.16	0.15	99.08	98.56	1.65	1.48	0.05	0.05
REDI-Face(ours)	RN34	100	92.42	91.78	88.69	1.96	2.50	2.72	0.14	0.17	95.48	97.23	***0.04***	1.93	0.16	0.11
REDI-Face(ours)	IR18	100	93.39	92.94	90.51	1.65	1.82	2.38	0.13	0.14	96.68	98.06	1.22	2.95	0.14	*0.08*
REDI-Face(ours)	RN50	100	*94.99*	*94.09*	91.62	1.48	1.69	2.20	*0.09*	0.11	*97.51*	98.30	0.18	1.70	***0.12***	*0.08*
REDI-Face(ours)	ViT-T	100	94.72	94.01	*91.67*	*1.33*	*1.44*	*2.14*	0.10	*0.10*	97.09	*98.38*	0.29	1.80	0.14	*0.08*
REDI-Face(ours)	RN101	100	95.00	94.09	91.31	1.45	1.79	2.27	0.09	0.11	97.61	98.35	0.55	1.57	0.13	0.08
REDI-Face(ours)	IR50	100	94.98	**95.21**	**92.5**	**1.19**	1.28	2.16	**0.08**	**0.09**	97.74	98.59	0.40	1.91	**0.12**	**0.07**
REDI-Face(ours)	ViT-S	100	95.48	94.77	92.06	1.21	**1.27**	2.05	0.11	**0.09**	97.33	98.53	0.42	1.84	0.13	**0.07**
REDI-Face(ours)$_{pt}$	IR18	100	94.49	94.00	91.85	1.39	1.53	2.10	0.13	0.14	97.09	98.39	0.19	2.00	0.13	0.08
REDI-Face(ours)$_{pt}$	IR50	100	**95.55**	95.01	92.36	1.26	1.42	**2.02**	0.18	0.16	**97.83**	**98.72**	0.6	1.65	0.13	0.08

Table 1. Comparison of demographic attribute inference pipelines based on accuracy, fairness, and robustness dimensions. Baselines are color-coded in red while our REDI-Face pretrained here with 1M WebFace images is highlighted in green with backbones similar to that of Fairface, in blue for larger ones and i norange when pretrained with 40M WebFace images [68] ($_{pt}$ suffix). We report the architecture of the models backbone (col. 2) and the number of training images annotated with regard to ethnicity/gender (col. 3). FairFace$_R$ and FairFace$_B$ indicate unsuccessful attempts to reproduce FairFace with their public dataset and the BUPT 100k training subset. The best results for compact models (red and green rows) are in *italics*, those including larger ones (red, green, and blue rows) are underlined and global best results are in **bold**. Higher is better for accuracy, while lower is better for fairness and robustness.

Arch.	**BFW**		**RFW**		**BUPT**	
	Imb	Bal	Imb	Bal	Imbb	Bal
IR-18	92.7	**93.2**	92.6	**93.0**	**90.5**	90.4
IR-50	95.3	**95.6**	94.8	**95.2**	91.8	**92.5**
RN101	**94.0**	93.8	**93.4**	93.4	90.8	**91.1**
RN34	89.8	**91.4**	88.4	**90.5**	86.9	**87.9**
RN50	94.1	**95.0**	93.5	**94.7**	90.8	**91.6**
ViT-s	94.5	**95.5**	94.0	**94.8**	91.4	**92.1**
ViT-t	93.7	**94.7**	93.4	**94.0**	91.3	**91.7**

Table 2. BFW, RFW, and BUPT accuracy when pretraining the IR50 backbone with Imbalanced (Imb) or balanced (Bal) WebFace subset comprising 1M images.

	1	**2**	**5**	**10**	**15**	**20**	**40**
500	85.2	87.3	88.4	89.4	89.2	89.5	89.6
1000	87.8	89.8	90.3	90.6	90.8	90.8	91.0
2000	89.6	90.2	90.8	91.3	91.6	91.4	92.1
5000	90.9	91.6	91.8	93.7	94.2	94.3	—
10000	91.7	91.5	94.3	94.8	—	—	—
20000	92.0	94.3	95.2	—	—	—	—

Table 3. Impact of the structure of the ethnicity classification dataset. We vary the number of identities (rows) and the number of samples per identity (columns). 40 is the maximum number of samples per identity. We limit the total dataset size to 100k. We report the RFW accuracy obtained with the IR50 backbone pretrained on 1M WebFace images.

structures of the BUPT subset used to train the ethnicity classification head. The results on the individual rows and columns indicate that a larger training dataset usually improves accuracy. Going beyond 100,000 training samples would probably bring further slight improvement, but such an experiment is outside the immediate scope of our study. The comparison of identity-sample combinations totaling the same number of images indicates a systematic advantage for configurations including more identities and a lower sample count per identity. This finding hints at the usefulness of maximizing diversity in DAI training datasets, to limit the number of sensitive attributes they contain.

Inference robustness detail. The robustness per ethnicity (Fig. 4) highlight a far higher stability (both in terms of total robustness and group wise robustness variance) for REDI-Face compared with the three ethnicity-based baselines, DeepFace FaceXFormer and FairFace. Moreover, the definition of the robustness metrics allows to compare with non related ethnic taxonomy such as the skin color [60]. The robustness metrics allows us to put in perspective the limitations of this approach, which exhibits poor robustness globally, and scores vary significantly across segments. This last comparison exemplifies the applicability of the robustness metric introduced in Eq. (6) to different segmentation schemes and questions the practical utility of the metric introduced in [60] since it is unable to predict the same segment for different photos of the same identity.

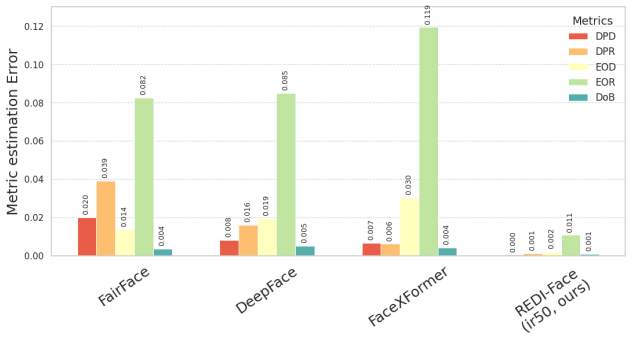


Figure 3. Absolute error between the fairness metrics on RFW and the ones estimated with annotators when considering a FAS trained on Casia.

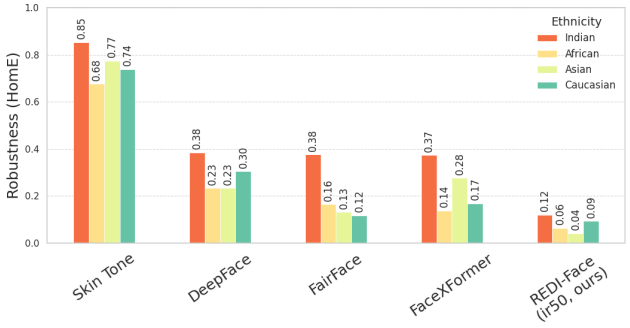


Figure 4. Ethnicity segments robustness (lower is better) in BFW for the three main baselines, REDI-Face pretrained with 1M WebFace images, and with four skin-color-based segments from [60].

5. Limitations and Ethics

Ethnicity segmentation is coarse and limited. The use of predefined ethnicity categories simplifies the inference but fails to capture the complex, continuous, and intersectional nature of human phenotypic diversity. In this work, we adopt a four-class scheme (Asian, Black, Indian, White). While commonly used and empirically robust, it reduces rich demographic variation into coarse groupings. This simplification may obscure subtle patterns of bias or robustness within subgroups and may limit generalizability across cultural or geographic contexts. Designing segmentation schemes that are both ethically meaningful and statistically reliable remains an open challenge, requiring careful tradeoffs between interpretability, granularity, and practicality.

Model size and pretraining data: larger may still be better. Our results show that transfer learning from compact face recognition models yields strong performance. However, we also observe modest but consistent gains when using larger architectures and more extensive pretraining datasets. This suggests that current performance may be suboptimal and that further scaling could improve outcomes. Nonetheless, training or fine-tuning large models introduces computational and environmental costs, and large-scale datasets often lack transparency regarding their demographic composition. Balancing scalability, efficiency, and dataset accountability is increasingly important in the era of foundation models.

Limited interpretability of learned demographic features. While our approach improves performance across accuracy, fairness, and robustness, the internal representations learned by deep models remain largely opaque. This lack of interpretability hampers transparency in high-stakes applications, making it difficult to audit which features influence predictions. Improving interpretability without sacrificing accuracy remains an important research direction, particularly as current architectures prioritize representational abstraction over explainability.

Ethical risks and potential misuse. Although our models are designed for fairness auditing and scientific reproducibility, they could be repurposed for unethical uses such as surveillance, profiling, or discrimination. The ability to infer sensitive demographic attributes from facial images at scale raises significant concerns around consent, privacy, and bias amplification. While releasing models and code promotes transparency and peer scrutiny, technical safeguards alone are insufficient. A broader dialogue is needed within the research, ethics, and policy communities to define acceptable use cases, auditing frameworks, and accountability mechanisms.

6. Conclusion

We revisited the DAI process and proposed a transfer learning–based formulation, supported by a theoretical motivation for improving the reliability of DAI pipelines. We extended the evaluation framework beyond accuracy and fairness by introducing robustness as a third critical dimension, and proposed HomE, an entropy-based robustness metric grounded in intra-identity consistency. We implemented and evaluated the proposed approach for both ethnicity and gender inference, demonstrating consistent performance gains—particularly for ethnicity, which remains the more challenging attribute. These results were obtained with a simple yet effective probing-based design, showing that a modular and interpretable architecture can match or surpass existing end-to-end systems.

Looking forward, we will address the limitations identified above. First, refining ethnicity categorization remains an open challenge, as shown by the limited robustness of recent schemes such as [60]. We will explore the construction of datasets organized around finer-grained ethnic categories with coherent phenotypic traits, as well as unsupervised, data-driven segmentations based on clustering embeddings from large pretrained models. Second, we plan to incorporate new foundation models as they emerge, eval-

uating their impact on DAI performance and robustness. Third, we will investigate techniques for interpreting demographic embeddings, such as concept activation vectors and disentanglement-based methods.

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A. Implementation detailled description

A.1. Datasets

FairFace [26] is described in detail in section 2.

BUPT [63] is a dataset of real images that can be found at this URL. We actually consider *BUPT-BalanceFace* that contains 1.3M images from 28k celebrities. Four labels are considered, namely *caucasian*, *indian*, *asian* and *african*. The dataset is balanced w.r.t the number of identity per class (7k per label) and approximately balanced in terms of number of images: 324k for *african*, 325k for *asian*, 326k for *caucasian* and 375k for *indian*.

WebFace [68] is a dataset of real images, containing 4M identities represented by 260M face images, which a part (4M identities and 42M face images) has been curated to result into Webface42M.

A.2. Training details

For training the end-to-end classifier (i.e., FairFace$_R$ and FairFace$_B$), we set the training duration to 30 epochs with a batch size of 64. We use the ADAM optimizer coupled with a learning rate at 10e-4 following the instructions of the original FairFace authors. For data augmentation during training, we applied random horizontal flipping, low-resolution augmentation, and random centered cropping with the same parameters as those in the AdaFace [28] repository. The CrossEntropy loss converges to 0.03 at the end of training, demonstrating successful convergence. See subsection A.4 for reproductivity details.

For training the **face recognition classifiers**, we employed the standard setup from AdaFace. We used the AdaFace loss with parameter h set to 0.333 and m set to 0.4. We applied the standard framework augmentations for face recognition training: random flipping, crop augmentation, low-resolution augmentation, and photometric augmentation. We trained the model for 30 epochs using the SGD optimizer with momentum and weight decay fixed at 5e-4. The scheduler was a step LR scheduler that divides the learning rate by 10 at epochs 12, 20, and 24. To ensure stability of the AdaFace loss when training the ResNets (torchvision implementation), we add a fully connected layer to the models to project the original 2024 latent space to 512.

The SVM implementation is from scikit-learn [43]. To accelerate the computation of cross distances, we implemented a wrapper that performs this phase on GPU. This allows us to drastically reduce both training and inference time (training time drops from 40 minutes to 2 minutes on a training set of 100k samples). For the RBF kernel, we use the scaling gamma defined as $\frac{1}{D \cdot \mathrm{Var}(X)}$, where D is the dimension of the feature vectors (512) and X are the input vectors. Since BUPT is balanced with regard to ethnicity but not gender, we use balanced weighting for gender defined as: $\mathrm{Weight}(\mathrm{Class}_i) = \frac{0.5}{\mathrm{Frequency}(\mathrm{Class}_i)}$. We use the squared hinge loss with L2 penalization and 1.0 as the regularization parameter.

All trainings are performed on a compute device having 250 GB of RAM, 96 CPU cores and an NVIDIA H100 GPU.

A.3. Architecture Parameters

Architecture(s)	#Parameters (M)	GFLOPS
ViT-T	19.1	2.76
RN34	21.3	1.94
IR18	24.0	5.24
RN50	24.6	2.18
IR50	43.5	12.64
RN101	42.5	4.04
ViT-S	76.0	11.00
Swin-B	109	30.92
VGG-Face	134	31.06
QWEN3 4B	4437	1101
QWEN3 8B	8767	1973

Table 4. Parameter count and FLOPS (computed with ptflops) to architecture mapping for DAI pipelines.

We report in Table 4 the parameters and GFLOPS associated with each architecture mentioned in the main paper. While allowing for significantly better performances, the backbones used by REDI-Face require lower parameter count and computational power.

A.4. Extended FairFace Reproducibility Analysis

In this section, we provide a forensic analysis of the widely used FairFace baseline, especially in Face recognition setups, and which performs the best across the considered baselines. While the original authors published weights report an accuracy of roughly 88% on the RFW dataset, the absence of public training code and the ambiguity regarding the exact training subset used have led to reproducibility challenges, as highlighted in the main paper. We investigate these discrepancies through two lenses: a weight distance analysis of the released model and an extensive hyperparameter optimization search. The objective of this section is two folds: verifying that the obtained reproduction results in Table 1 (i.e. for FairFace$_R$ and FairFace$_B$) are not due to a poor choice of hyperparameters, and justifying the choice of BUPT for the training of our approach, REDI-Face.

A.4.1. Forensic Weight Analysis

The training details in [26] are sparse as the authors only mention the learning rate and the optimizer they used. However, examination of the publicly available FairFace implementation reveals several code patterns and comments suggesting the use of ImageNet-pretrained ResNet-34 weights

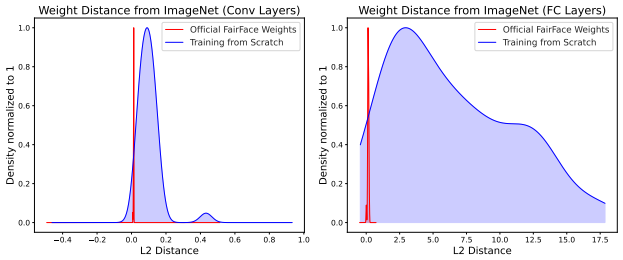


Figure 5. Distribution of the distances for the model release by FairFace authors and for a model trained on FairFace from scratch

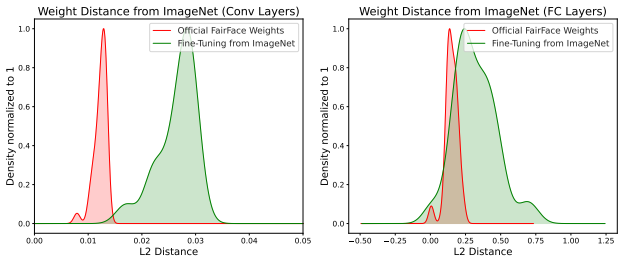


Figure 6. Distribution of the distances for the model release by FairFace authors and for a model trained on FairFace initialized with the ImageNet torchvision checkpoint

at training initialization, although not specified in the paper. To infer the training dynamics used for the released FairFace model, and understand the reproduction attempt performances in Table 1, we first analyzed the Euclidean distance between the weights of the released FairFace ResNet34 model and the standard torchvision ImageNet-pretrained weights. We compared these distances against two control models trained by us: one trained from scratch and one fine-tuned from ImageNet.Hypothesis: If the authors trained from scratch, the weights should be uncorrelated with ImageNet.

As illustrated in Figure Figure 6 and Figure 5, we observe a distinct trend. The weights of the official FairFace model exhibit a weight distribution that is significantly closer to the ImageNet initialization, not only when compared to the model trained from scratch, but also to a certain extent, closer to our own fine-tuned models. This proximity suggests that the official FairFace model might have been initialized with ImageNet weights and might also have been trained with extremely conservative hyperparameters (such as a very low learning rate or some weight decay).

A.4.2. Hyperparameter Optimization

To rigorously rule out suboptimal hyperparameter choices as the cause for the performance gap, we conducted an extended search using the Optuna framework. We performed two distinct experiments, optimizing for accuracy on the RFW dataset, and averaged each trial over three random

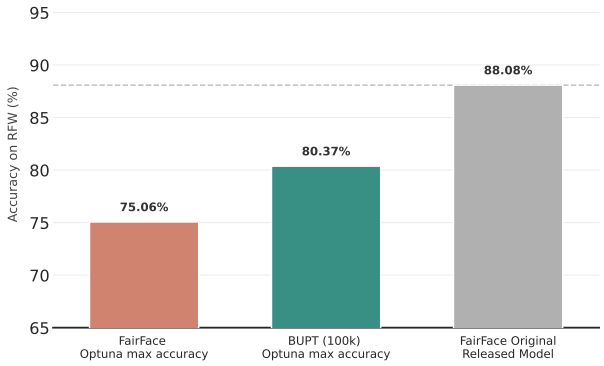


Figure 7. Comparison between the performance on RFW using the released model from FairFace authors [26] and the max accuracy obtained with Optuna, training on either the FairFace dataset or 100k images from the BUPT dataset

seeds. Optimizing on the test set is obviously not a good practice when reporting model performances, but our purpose here is to define an upper bound for the achievable performances for the considered training set.

We explored a wide range of configurations :

- Starting from Image Net torchvision checkpoint, or learning from scratch
- ADAM or SGD optimizer
- OneCycleLR scheduler or None
- Learning rate choice from 10e-6 to 10e-1
- With or without weight decay
- Choice of the number of epochs from 1 to 30

After 160 trials, the best performance achievable using the public FairFace training data was 75.06%, significantly trailing the 88% obtained with the released weights. Conversely, applying the same optimization process to a subset of the BUPT dataset of identical size (100k images) yielded 80.37%. For both training sets, better performances were obtained when starting the training from the torchvision ImageNet weights. For the training on the FairFace dataset however, the weight decay was detected by Optuna as being a variable with a positive effect, which wasn't the case for the training on BUPT, showing that training on FairFace needs some extra regularization.

Training Source	Best Accuracy (RFW)	Key Hyperparameters (Best Trial)
FairFace	75.06%	LR: 7.43×10^{-5}, Adam, Pretrain: True, Epochs: 29
BUPT	**80.37%**	LR: 3.26×10^{-5}, Adam, Pretrain: True, Epochs: 25
FairFace (Released)	*88.08%*	*Unknown / Not reproducible*

Table 5. Optimal Optuna Hyperparameters for training on FairFace and on a subest of BUPT

A.4.3. Reproduction Attempts Summary

The discrepancy between the optimized result (75.06%) and the reported result (88.08%) is too large to be attributed to hyperparameter noise. This suggests that the publicly released FairFace training set is likely not the same dataset (or at least not of the same quality) as the one used to generate the released model weights.

These findings underscore the reproducibility challenges inherent in opaque, end-to-end training pipelines where data provenance is unclear. This is not the case solely for FairFace, but also for DeepFace [54] for which training details are severely lacking, and FaceXFormer, whose training scripts have yet to be released.

This analysis serves as a strong motivation for advocating simpler, modular solutions - such as the transfer learning approach proposed in this work - which not only yields superior performance but also guaranties the transparency essential for rigorous academic research.

Finally, given the significant gap in achievable performance (80.37% vs. 75.06%) under identical constraints, we conclude that the BUPT training set represents a more robust and reliable candidate for ethnicity classification than the publicly available FairFace data.

B. Details on metrics

B.1. Performance Metrics

The face recognition performance is estimated with *Micro-Average Accuracy* which is reelvant when dealing with unbalanced data, as it gives equal importance to each dataset segment, regardless its size. Therefore, the metrics is not biased toward the majority group. Other possible performance metrics include:

- *True Match Rate (TMR)* or *True positive Rate (TPR)* or *Sensitivity* is the number of actual positive cases correctly identified over the actual number of positive samples in the dataset
- *False Match Rate (FMR)*, or *False Positive Rate (FPR)*, is the number of negative cases incorrectly identified as positive by the FAS over the actual number of negative samples.

B.2. Fairness Metrics

To estiamte the fairness in the main paper, we considered the Degree of Bias (DoB) [16] which is the standard deviation of the per-group accuracies. A lower standard deviation means that the performance varies less from one group to another, thus that the FAS is fairer.

Demographic Parity Difference (DPD) and *Demographic Parity Ratio (DPR)* [1, 2] are derived from the same principle, aiming at achieving equal representation of the demographic groups. To compute them, one consider the lowest p_{min} and highest p_{max} probability across all subgroups then $DPD = |p_{max} - p_{min}|$ and $DPR = \frac{p_{min}}{p_{max}}$. Hence both are scores in $[0, 1]$ but the lower DPD, the faire the FAS while it is fairer when DPR is closer to one.

Equalized Odds Difference (EOD) and *Equalized Odds Ratio (EOR)* [1, 21] require that the FAS's performances are independent of the demographic group, that is the TPR or FPR are the same across all groups. The EOD is defined by the absolute difference between smallest and largest values for the considered performance metric (TPR or FPR) across groups, while EOR is defined according to their ratio. Similarly to demographic parity, these matrics are in $[0, 1]$ a FAS is perfectly fair when its EOD is zero or its EOR is one.

We present a formal presentation of these metrics and their link to the binary indicator of performance (Y_i in the main article) in subsection C.4.

B.3. Label Robustness Metric

Let us consider an image dataset $\mathcal{I} = \{I_j^{id}; id \in [\![1, D]\!], j \in [\![1, N_{id}]\!]\}$, with D identities, an identitity id being represented by N_{id} images. Such dataset are typically BFW, BUPT, CASIA, or GLINT. We consider a labeling function $f_{\mathcal{A}}^{\mathcal{C}}$ that attributes a label to each image, each label being an element of a finite set $\mathcal{C}$, and depends of a dataset $\mathcal{A}$ ($f_{\mathcal{A}}^{\mathcal{C}}$ is defined below). We propose to consider three metrics to reflect the robustness of the labeling system $(\mathcal{A}, \mathcal{C})$. The general idea is that a good labeling system should (at least) attribute as much as possible images of a given identity to the same label.

Macro majority accuracy. For each identity id, we consider the frequency of the majority label and compute the average over all labels of the dataset

$$MaMA = \frac{\sum_{id=1}^{D} \max_{k=1\dots N_{id}} \left(\sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})} \right)}{\sum_{id} N_{id}} \quad (7)$$

Micro majority accuracy For each identity, we consider the proportion of images labeled with the majority label and compute its average over all identities:

$$MiMA = \frac{1}{D} \sum_{id=1}^{D} \frac{1}{N_{id}} \max_{k=1\dots N_{id}} \left(\sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})} \right) \quad (8)$$

Last, let remind that in the main paper we consider, for each identity id, the relative frequency of each possible label $k \in \mathcal{C}$ noted as $p_{id}^k = \frac{1}{N_{id}} \sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})}$. We then compute the **homogeneity entropy** as the average entropy of p over all identities:

$$HomE = \frac{1}{D} \sum_{id} \frac{1}{\log(C)} \sum_{k} -p_{id}^k \log(p_{id}^k) \quad \text{(6 revisited)}$$

These metrics reflect the average variation of the attributions for each entity. Ideally, all images of a given entity are attributed to the same label. In this case, the *homogeneity entropy macro majority accuracy* and *micro majority accuracy* are respectively 0,1 and 1. In the worst case, the images of a given entity are uniformly distributed over all C labels, leading to the three metrics above being respectively 1, $\frac{1}{C}$ and $\frac{1}{C}$. We thus normalize MiMA and MaMA by multiplying by $\frac{C}{C-1}$ and removing $\frac{1}{C-1}$ such that the worst value is 0 and the best value is 1

Labelling function the function $f_{\mathcal{A}}^{\mathcal{C}}$ project an image onto a unique label in $\mathcal{C}$. It is defined according an image dataset $\mathcal{A}$. We consider two versions of $f_{\mathcal{A}}^{\mathcal{C}}$:

- a non-parametric function: each image of $\mathcal{A}$ is described with an image encoder that project it into an euclidean space. It then attributes a label to an image according to its nearest neighbors (1-NN or k-NN)
- a parametric function, such as a neural network, that is learnt on $\mathcal{A}$, with labels in $\mathcal{C}$, and a cross-entropy loss.

B.4. Robustness Results

Results of robustness for all the methods are reported in Table 6, in term of all three metrics. We also report, similarly as Figure 4, the detailed ethnicity-wise robustness results on BUPT in Figure 8. The results are consistent with the ones reported in Figure 4, highlighting the stability of our pipeline REDI-Face and supporting the argumentation in the main paper.

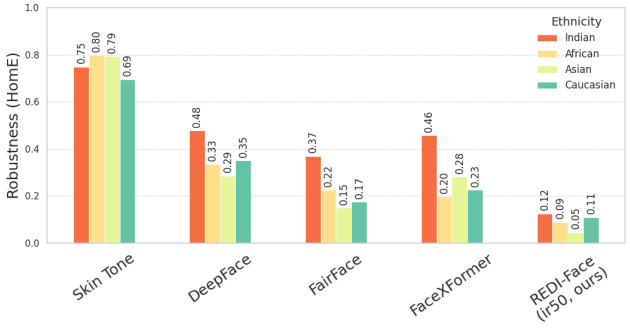


Figure 8. Ethnicity segments robustness (lower is better) on BUPT for the three main baselines, REDI-Face pretrained with 1M WebFace images, and with four skin-color-based segments from [60]

C. Details on Theoretical Aspects

C.1. Theoretical Justification for RBF-SVM on Angular Margin Embeddings

We provide a novel unified geometric and probabilistic framework that theoretically justifies the use of RBF kernel classifiers for transfer learning from face recognition models pretrained with angular margin losses to demographic attribute inference. While the individual mathematical components we build upon are established results from directional statistics [35] and metric learning [11, 53], **our contribution is the first end-to-end theoretical analysis connecting (i) vMF-structured pretraining, (ii) vMF kernel similarity, and (iii) vMF mixture decision boundaries in a unified framework for this transfer learning scenario**. Specifically, we characterize RBF-SVM decision functions as comparisons between von Mises-Fisher mixture models—an interpretation that has not appeared in prior work and provides principled guidance for architectural choices for DAI in fairness-aware face recognition.

C.1.1. Background: Angular Margin Losses and Hyperspherical Geometry

Modern face recognition models employ angular margin losses such as ArcFace [11] or AdaFace [28], which operate on L2-normalized feature embeddings. Let $f : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^d$ denote a deep neural network that maps an input face image to a d-dimensional embedding. The normalization constraint ensures that feature vectors lie on the unit hypersphere:

$$\mathbf{z} = \frac{f(\mathbf{x})}{\|f(\mathbf{x})\|_2} \in \mathbb{S}^{d-1} := \{\mathbf{v} \in \mathbb{R}^d : \|\mathbf{v}\|_2 = 1\}. \quad (9)$$

The ArcFace loss for a training sample $(\mathbf{x}_i, y_i)$ with ground-truth identity $y_i \in \{1, \dots, C\}$ is:

$$\mathcal{L}_{\text{ArcFace}} = -\log \frac{\exp(s\cos(\theta_{y_i} + m))}{\exp(s\cos(\theta_{y_i} + m)) + \sum_{j\neq y_i} \exp(s\cos\theta_j)}, \quad (10)$$

where $\theta_j = \arccos(\mathbf{w}_j^\top \mathbf{z}_i)$ is the angle between the embedding $\mathbf{z}_i$ and the j-th class center $\mathbf{w}_j \in \mathbb{S}^{d-1}$, s is a scale parameter, and m is the angular margin penalty.

C.1.2. Established Connection: Angular Losses as von Mises-Fisher Distributions

The von Mises-Fisher (vMF) distribution is the canonical probability distribution for directional data on the unit hypersphere [35]. For a random unit vector $\mathbf{z} \in \mathbb{S}^{d-1}$, the vMF density is:

$$p(\mathbf{z} \mid \boldsymbol{\mu}, \kappa) = C_d(\kappa) \exp\left(\kappa \boldsymbol{\mu}^\top \mathbf{z}\right), \quad (11)$$

where $\boldsymbol{\mu} \in \mathbb{S}^{d-1}$ is the mean direction, $\kappa \geq 0$ is the concentration parameter, and $C_d(\kappa) = \kappa^{d/2-1}/[(2\pi)^{d/2} I_{d/2-1}(\kappa)]$ is the normalization constant with I_ν denoting the modified Bessel function of the first kind.

It is well-known [35, 53] that the softmax classifier with L2-normalized features and class centers is equivalent to a generative model where each class follows a vMF distribution. Specifically, under uniform class priors, the softmax probability (without angular margin) can be derived

DAI Pipeline	Arch.	#labels (K)	Ethnicity						Gender					
			MaMA ↑		MiMA ↑		HomE ↓		MaMA ↑		MiMA ↑		HomE ↓	
			BFW	BUPT	BFW	BUPT	BFW	BUPT	BFW	BUPT	BFW	BUPT	BFW	BUPT
FairFace [26]	RN34	89	0.90	0.88	0.90	0.88	0.19	0.22	0.96	***0.98***	0.96	0.97	0.19	0.22
FairFace (reprod)	RN34	89	0.80	0.79	0.80	0.79	0.38	0.41	0.80	0.79	0.80	0.79	0.25	0.19
BUPT (reprod)	RN34	89	0.80	0.90	0.80	0.89	0.35	0.18	0.94	0.97	0.95	0.96	0.19	0.12
DeepFace[54]	VGG-Face	?	0.84	0.80	0.84	0.80	0.29	0.36	0.90	0.94	0.90	0.95	0.31	0.16
FaceXFormer	Swin-B	≈ 150	0.86	0.84	0.86	0.84	0.0.23	0.26	0.96	0.97	0.95	0.96	0.12	0.08
RN34+RBF	RN34	100	0.93	0.91	0.93	0.90	0.14	0.17	0.96	***0.98***	0.96	0.97	0.16	0.11
IR18+RBF	IR18	100	0.94	0.92	0.94	0.93	0.13	0.14	0.96	***0.98***	***0.97***	***0.98***	0.14	*0.08*
RN50+RBF	RN50	100	*0.95*	*0.94*	*0.95*	*0.94*	*0.09*	0.11	***0.97***	***0.98***	***0.97***	***0.98***	***0.12***	*0.08*
ViT-T+RBF	ViT-T	100	*0.95*	0.93	*0.95*	*0.94*	0.10	*0.10*	0.96	***0.98***	***0.97***	***0.98***	0.14	*0.08*
RN101+RBF	RN101	100	**0.96**	0.94	0.96	0.94	0.09	0.11	**0.97**	**0.98**	**0.97**	**0.98**	0.13	0.08
IR50+RBF	IR50	100	**0.96**	**0.95**	**0.96**	**0.95**	**0.08**	**0.09**	0.96	**0.98**	**0.97**	**0.98**	**0.12**	**0.07**
ViT-S+RBF	ViT-S	100	**0.96**	0.94	**0.96**	**0.95**	0.11	**0.09**	**0.97**	**0.98**	**0.97**	**0.98**	0.13	**0.07**
IR18+RBF	IR18	100	0.93	0.92	0.94	0.93	0.13	0.14	**0.97**	**0.98**	**0.97**	**0.98**	0.13	0.08
IR50+RBF	IR50	100	0.91	0.92	0.91	0.92	0.18	0.16	0.96	**0.98**	**0.97**	**0.98**	0.13	0.08

Table 6. Results with all robustness metrics. Column 3 (#imgs) is the number of images used for pretraining (in millions). See Table 4 for details on models. The best results for compact models comparable with FairFace [26] (red and green rows) are in *italics*. The best results obtained when including larger pretrained architectures (red, green, and blue rows) are underlined. The best global results in **bold**. Higher figures are better for accuracy, while lower figures are better for fairness and robustness.

DAI Pipeline	Arch	#labels (K)	Ethnicity								Gender					
			Accuracy ↑			DAI Fairness ↓			Robustness ↓		Accuracy ↑		DAI Fairness ↓		Robustness ↓	
			BFW	RFW	BUPT	BFW	RFW	BUPT	BFW	BUPT	BFW	BUPT	BFW	BUPT	BFW	BUPT
REDI-Face(linear)	RN34	100	90.12	89.51	86.58	2.42	2.88	2.90	0.20	0.22	94.21	96.44	0.86	1.30	0.19	0.15
REDI-Face(linear)	IR18	100	89.69	89.78	87.81	1.71	1.57	2.45	0.20	0.20	94.48	96.46	1.58	0.07	0.18	0.15
REDI-Face(linear)	RN50	100	89.04	87.77	87.07	1.60	1.63	2.76	0.23	0.21	93.66	96.04	1.35	1.13	0.19	0.16
REDI-Face(linear)	ViT-T	100	90.52	90.61	88.48	2.05	2.34	2.66	0.19	0.19	94.51	96.54	2.55	0.26	0.19	0.14
REDI-Face(linear)	RN101	100	88.20	87.24	87.14	1.39	1.52	2.56	0.25	0.22	93.58	95.84	2.50	0.63	0.20	0.17
REDI-Face(linear)	IR50	100	90.61	90.65	87.95	2.14	2.58	2.68	0.19	0.20	94.96	96.85	1.55	1.32	0.17	0.13
REDI-Face(linear)	ViT-S	100	90.75	90.96	88.54	1.91	2.17	2.67	0.18	0.18	94.43	96.53	2.05	0.01	0.19	0.15

via Bayes' theorem:

$$P(y = c \mid \mathbf{z}) = \frac{p(\mathbf{z} \mid \mathbf{w}_c, s)}{\sum_{j=1}^{C} p(\mathbf{z} \mid \mathbf{w}_j, s)} = \frac{\exp(s\mathbf{w}_c^\top \mathbf{z})}{\sum_{j=1}^{C} \exp(s\mathbf{w}_j^\top \mathbf{z})}, \tag{12}$$

where each class c is modeled as a vMF distribution with mean direction $\mathbf{w}_c$ and shared concentration parameter $\kappa = s$. The normalization constants $C_d(s)$ cancel in the ratio.

The angular margin m in ArcFace modifies this by effectively decreasing the likelihood of the correct class during training through the transformation $\cos(\theta_c) \to \cos(\theta_c + m)$, forcing the model to achieve higher concentration (smaller intra-class angles) to overcome the penalty. This results in compact, well-separated vMF-like clusters on $\mathbb{S}^{d-1}$ after convergence.

C.1.3. Established Connection: RBF Kernel as von Mises-Fisher Similarity

The standard RBF (Gaussian) kernel between two vectors is:

$$K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}_j) = \exp\left(-\gamma \|\mathbf{z}_i - \mathbf{z}_j\|_2^2\right). \tag{13}$$

For L2-normalized vectors $\mathbf{z}_i, \mathbf{z}_j \in \mathbb{S}^{d-1}$, a standard calculation yields:

$$\|\mathbf{z}_i - \mathbf{z}_j\|_2^2 = \mathbf{z}_i^\top \mathbf{z}_i - 2\mathbf{z}_i^\top \mathbf{z}_j + \mathbf{z}_j^\top \mathbf{z}_j = 2(1 - \mathbf{z}_i^\top \mathbf{z}_j). \tag{14}$$

Substituting into Eq. (13) and defining $\kappa_{\text{eff}} := 2\gamma$:

$$K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}_j) = \exp(-\kappa_{\text{eff}}) \cdot \exp(\kappa_{\text{eff}} \mathbf{z}_i^\top \mathbf{z}_j). \tag{15}$$

From the vMF density Eq. (11), setting $\boldsymbol{\mu} = \mathbf{z}_i$:

$$p(\mathbf{z}_j \mid \mathbf{z}_i, \kappa_{\text{eff}}) = C_d(\kappa_{\text{eff}}) \exp(\kappa_{\text{eff}} \mathbf{z}_i^\top \mathbf{z}_j). \tag{16}$$

Therefore, the RBF kernel satisfies:

$$K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}_j) = \frac{\exp(-\kappa_{\text{eff}})}{C_d(\kappa_{\text{eff}})} \cdot p(\mathbf{z}_j \mid \mathbf{z}_i, \kappa_{\text{eff}}) \propto p(\mathbf{z}_j \mid \mathbf{z}_i, \kappa_{\text{eff}}). \tag{17}$$

This establishes a classical result from directional statistics [35]: **the RBF kernel on normalized features computes a similarity measure proportional to the vMF likelihood**. The constant of proportionality does not affect the SVM decision boundary, as it scales all kernel evaluations uniformly.

C.1.4. Novel Contribution: Decision Boundaries as von Mises-Fisher Mixture Comparisons

We now present our key theoretical contribution: characterizing the RBF-SVM decision function as a comparison between vMF mixture models. This interpretation has not appeared in prior work and provides new insight into why RBF kernels are theoretically suited for this transfer learning scenario.

Consider binary demographic classification where $y_i \in \{-1, +1\}$ indicates class membership. The SVM decision function is:

$$f(\mathbf{z}) = \sum_{i \in \mathcal{S}} \alpha_i y_i K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}) + b, \tag{18}$$

where $\mathcal{S}$ is the set of support vector indices and $\alpha_i > 0$ are the dual coefficients.

Partitioning support vectors by class: $\mathcal{S}_+ := \{i \in \mathcal{S} : y_i = +1\}$ and $\mathcal{S}_- := \{i \in \mathcal{S} : y_i = -1\}$, we rewrite:

$$f(\mathbf{z}) = \sum_{i \in \mathcal{S}_+} \alpha_i K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}) - \sum_{i \in \mathcal{S}_-} \alpha_i K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}) + b. \tag{19}$$

Substituting the vMF equivalence from Eq. (17):

$$f(\mathbf{z}) = \Lambda \left[\sum_{i \in \mathcal{S}_+} \alpha_i p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}) - \sum_{i \in \mathcal{S}_-} \alpha_i p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}) \right] + b, \tag{20}$$

where $\Lambda := \exp(-\kappa_{\text{eff}})/C_d(\kappa_{\text{eff}})$ is a positive constant.

Define normalized mixture weights:

$$w_i^+ := \frac{\alpha_i}{\sum_{j \in \mathcal{S}_+} \alpha_j} \quad \text{for } i \in \mathcal{S}_+, \quad \text{with} \sum_{i \in \mathcal{S}_+} w_i^+ = 1, \tag{21}$$

$$w_i^- := \frac{\alpha_i}{\sum_{j \in \mathcal{S}_-} \alpha_j} \quad \text{for } i \in \mathcal{S}_-, \quad \text{with} \sum_{i \in \mathcal{S}_-} w_i^- = 1. \tag{22}$$

Each sum in Eq. (20) can be interpreted as a **vMF mixture density**:

$$p_+(\mathbf{z}) := \sum_{i \in \mathcal{S}_+} w_i^+ p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}), \tag{23}$$

$$p_-(\mathbf{z}) := \sum_{i \in \mathcal{S}_-} w_i^- p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}). \tag{24}$$

These are well-defined mixture models where each support vector $\mathbf{z}_i$ serves as a vMF component center with concentration $\kappa_{\text{eff}} = 2\gamma$, and the normalized SVM coefficients $w_i^\pm$ act as mixture weights. The decision function becomes:

$$f(\mathbf{z}) = \Lambda \left[\beta_+ p_+(\mathbf{z}) - \beta_- p_-(\mathbf{z}) \right] + b, \tag{25}$$

where $\beta_\pm := \sum_{j \in \mathcal{S}_\pm} \alpha_j$ represent the total mass assigned to each class.

Log-likelihood ratio interpretation. The classification decision $\text{sign}(f(\mathbf{z}))$ is equivalent to testing:

$$\hat{y} = \text{sign} \left[\beta_+ p_+(\mathbf{z}) - \beta_- p_-(\mathbf{z}) + b/\Lambda \right]. \tag{26}$$

Under balanced support vector contributions ($\beta_+ \approx \beta_-$) and absorbing the bias into a threshold, this approximates a **log-likelihood ratio test**:

$$\hat{y} = \text{sign} \left[\log \frac{p_+(\mathbf{z})}{p_-(\mathbf{z})} \right] = \text{sign} \left[\log p_+(\mathbf{z}) - \log p_-(\mathbf{z}) \right], \tag{27}$$

which is the optimal Bayes classifier for choosing between two generative models.

C.1.5. Implications: Geometric Consistency for Transfer Learning

This theoretical analysis reveals a fundamental geometric consistency in our transfer learning pipeline that has not been previously formalized:

1. **Pretraining with angular margin losses** (e.g., AdaFace) organizes face embeddings on the hypersphere such that same-identity samples form tight vMF-like clusters (high κ), while different identities are angularly separated—this is established [11, 53].
2. **RBF kernel SVMs** operating on these normalized embeddings implicitly leverage the vMF geometry through Eq. (17), providing non-linear decision boundaries that respect the spherical structure—the kernel-vMF connection is established [35], but its application to this transfer learning scenario is novel.
3. **Decision boundaries as vMF mixture comparisons** (Eqs. (23)–(27)) provide a new probabilistic interpretation: the RBF-SVM learns to discriminate between demographic attributes by comparing which vMF mixture (constructed from support vectors) better explains a test embedding—**this characterization is our novel contribution**.

Theoretical justification for RBF over linear classifiers. Unlike linear classifiers that assume Euclidean separability, the RBF kernel naturally accommodates the curved geometry of $\mathbb{S}^{d-1}$ through its vMF structure. While hyperspherical kernels have been studied for SVM [67], prior work did not connect them to vMF-based pretraining or characterize decision boundaries as mixture models. Our framework provides the first principled justification for RBF-SVM specifically in the context of transfer learning from angular-loss-pretrained face recognition to demographic inference.

Parameter selection guidance. The bandwidth parameter γ controls the effective concentration $\kappa_{\text{eff}} = 2\gamma$, allowing the classifier to adapt to the intrinsic clustering scale

induced during pretraining. Higher γ increases sensitivity to angular proximity, appropriate when pretraining has produced highly concentrated (small intra-class variance) clusters. This provides interpretable, geometry-motivated guidance for hyperparameter selection—a contrast to deep non-linear classifiers that introduce numerous architectural parameters without explicit geometric motivation.

C.2. Proofs of Propositions

Notations We consider a face analysis system (FAS) trained on an images dataset, each of these last being annotated by a class label (either binary on pairs of images for face verification or multicategorial for each image for face recognition) and a set of N_a sentitive attributes, each being categorial. The cartesian product of all possible classes of N_a attributes leads to classify the images into K groups[4].

Let $G \in \{1, \dots, K\}$ be the true latent group of an identity, $\widehat{G} \in \{1, \dots, K\}$ the estimated label of the corresponding sensitive attribute, and $Y \in \{0, 1\}$ a binary performance indicator associated to an identity for the considered FAS.

We define the prior of the sensitive attribute a as

$$\pi_a := \Pr(G = a) \tag{28}$$

and the probability of success of the FAS given the *true* said sensitive attribute as

$$p_a := \Pr(Y = 1 \mid G = a) \tag{29}$$

This last is actually estimated automatically with some attribute classifiers that have a confusion matrix $C = (c_{ab})_{a,b=1}^{K}$ with

$$c_{ab} := \Pr(\widehat{G} = b \mid G = a) \tag{30}$$

such that each row of C sums to 1. We then define:

$$\tau_b := \Pr(\widehat{G} = b) = \sum_{a=1}^{K} \pi_a c_{ab} \tag{31}$$

$$m_g := \Pr(Y = 1 \mid \widehat{G} = g) \tag{32}$$

For convenience of notations we set:

$$p = (p_1, \dots, p_K)^\top \tag{33}$$

$$\pi = (\pi_1, \dots, \pi_K)^\top \tag{34}$$

$$\tau = (\tau_1, \dots, \tau_K)^\top = C^\top \pi \tag{35}$$

and $\odot$ is the elementwise product. We consider an i.i.d. sample of I identities; for identity i we observe $(\widehat{G}_i, Y_i)$ but not G_i. For each observed label g we define the empirical plug-in estimator:

$$\widehat{m}_g = \frac{\frac{1}{I}\sum_{i=1}^{I} \mathbf{1}\{\widehat{G}_i = g\} Y_i}{\frac{1}{I}\sum_{i=1}^{I} \mathbf{1}\{\widehat{G}_i = g\}} \tag{36}$$

[4]for example, it there are 4 ethnicities, 2 genders and 4 classes of age, then $K = 4 * 2 * 4 = 32$

We note $\widehat{m} = (\widehat{m}_1, \dots, \widehat{m}_K)^\top$ assuming the denominator is nonzero for the K groups considered, and $\widehat{\tau}$ the empirical estimator of τ.

Proposition 1 (formal) If one assumes that $(Y \perp\!\!\!\perp \widehat{G}) \mid G$ (Equation 1) and $C^\top$ (thus C) is invertible, then:

- the empirical plug-in estimator $\widehat{m}_g$ converges in probability toward m_g as $I \to \infty$ and:

$$m_g = \frac{\sum_{a=1}^{K} \pi_a c_{ag} p_a}{\sum_{a=1}^{K} \pi_a c_{ag}} = \frac{1}{\tau_g} \sum_{a=1}^{K} \pi_a c_{ag} p_a \tag{37}$$

- the bias of the plug-in limit relative to the true group rate p_g satisfies:

$$\mathrm{Bias}_g := m_g - p_g = \frac{\sum_{a \neq g} \pi_a c_{ag} (p_a - p_g)}{\sum_{a=1}^{K} \pi_a c_{ag}} \tag{38}$$

and therefore if $c_{ag} \to 0$ for every $a \neq g$ and $c_{gg} \to 1$ then $\mathrm{Bias}_g \to 0$. Moreover, for $\Delta_{\max,g} := \max_{a \neq g} |p_a - p_g|$,

$$\left|\mathrm{Bias}_g\right| \leq \Delta_{\max,g} \cdot \frac{\sum_{a \neq g} \pi_a c_{ag}}{\pi_g c_{gg}} \tag{39}$$

Proof. For a fixed g, by the weak law of large numbers we have the convergence in probability of the numerator and denominator of Equation 36:

$$\frac{1}{I}\sum_{i=1}^{I} \mathbf{1}\{\widehat{G}_i = g\} Y_i \xrightarrow{p} \mathbb{E}[\mathbf{1}\{\widehat{G} = g\} Y] = \Pr(\widehat{G} = g,\, Y = 1) \tag{40}$$

and

$$\frac{1}{I}\sum_{i=1}^{I} \mathbf{1}\{\widehat{G}_i = g\} \xrightarrow{p} \Pr(\widehat{G} = g) = \tau_g \tag{41}$$

provided $\tau_g > 0$. Therefore the ratio $\widehat{m}_g$ converges in probability to

$$\frac{\Pr(\widehat{G} = g,\, Y = 1)}{\Pr(\widehat{G} = g)} = \Pr(Y = 1 \mid \widehat{G} = g) \tag{42}$$

that is m_g by definition. By the law of total probability and Equation 1 we have:

$$\Pr(\widehat{G} = g,\, Y = 1) = \sum_{a=1}^{K} \Pr(\widehat{G} = g,\, Y = 1,\, G = a)$$

$$= \sum_{a=1}^{K} \Pr(\widehat{G} = g \mid G = a) \Pr(Y = 1 \mid G = a) \Pr(G = a)$$

$$= \sum_{a=1}^{K} c_{ag}\, p_a\, \pi_a \tag{43}$$

Using Equation 31:

$$m_g = \frac{\sum_{a=1}^{K} \pi_a c_{ag} p_a}{\sum_{a=1}^{K} \pi_a c_{ag}} \tag{44}$$

that shows the first point of the proposition.

By definition,

$$\text{Bias}_g = m_g - p_g \tag{45}$$

$$= \frac{\sum_{a=1}^{K} \pi_a c_{ag} p_a}{\sum_{a=1}^{K} \pi_a c_{ag}} - p_g \tag{46}$$

$$= \frac{\sum_{a=1}^{K} \pi_a c_{ag} (p_a - p_g)}{\sum_{a=1}^{K} \pi_a c_{ag}} \tag{47}$$

$$\tag{48}$$

That is the same as Equation 38 since the term for $a = g$ vanished in the sum of the numerator. Applying the triangle inequality to the absolute value, we have:

$$|\text{Bias}_g| = \left| \frac{\sum_{a=1}^{K} \pi_a c_{ag} (p_a - p_g)}{\sum_{a=1}^{K} \pi_a c_{ag}} \right| \tag{49}$$

$$\leq \frac{\sum_{a=1}^{K} \pi_a c_{ag} |p_a - p_g|}{\sum_{a=1}^{K} \pi_a c_{ag}} \tag{50}$$

$$\leq \Delta_{\max,g} \cdot \frac{\sum_{a \neq g} \pi_a c_{ag}}{\sum_{a=1}^{K} \pi_a c_{ag}} \tag{51}$$

Since $\sum_{a=1}^{K} \pi_a c_{ag} \geq \pi_g c_{gg}$, we finally have the bound:

$$|\text{Bias}_g| \leq \Delta_{\max,g} \cdot \frac{\sum_{a \neq g} \pi_a c_{ag}}{\pi_g c_{gg}} \tag{52}$$

That is the same as Equation 39. Hence, if the confusion matrix tends towards identity then $c_{ag} \to 0$ for every $a \neq g$ and $c_{gg} \to 1$, thus the numerator $\sum_{a \neq g} \pi_a c_{ag} \to 0$ and denominator $\sum_a \pi_a c_{ag} \to \pi_g$, so $\text{Bias}_g \to 0$. □

Proposition 2 (formal) If $C^\top$ is invertible then the estimator

$$\widehat{p}^{(\text{corr})} = \frac{(C^\top)^{-1}(\widehat{\tau} \odot \widehat{m})}{\pi} \tag{53}$$

is asymptotically unbiased for p. For operator norms $\|\cdot\|_{\text{op}}$, its covariance matrix satisfies:

$$\| \text{Cov}\left((C^\top)^{-1}(\widehat{\tau} \odot \widehat{m})\right)\|_{\text{op}} \leq \|(C^\top)^{-1}\|_{\text{op}}^2 \cdot \| \text{Cov}(\widehat{\tau} \odot \widehat{m})\|_{\text{op}}. \tag{54}$$

Thus the factor $\|(C^\top)^{-1}\|_{\text{op}}^2$ controls the worst-case multiplicative increase in variance; when $C \to I$ the factor tends to 1, while ill-conditioned C make $\|(C^\top)^{-1}\|_{\text{op}}$ large and amplify variance.

Proof. Let us consider the population-observed numerator vector $n \in \mathbb{R}^K$ with $n_g := \Pr(\widehat{G} = g, Y = 1) = \sum_{a=1}^{K} \pi_a c_{ag} p_a$. We have $m_g = n_g / \tau$ and hence $n = \tau \odot m$. Thus:

$$n = C^\top (\pi \odot p) \tag{55}$$

$$\tau \odot m = C^\top (\pi \odot p) \tag{56}$$

Thus if $C^\top$ is invertible:

$$\pi \odot p = (C^\top)^{-1}(\tau \odot m) \tag{57}$$

$$p = \frac{(C^\top)^{-1}(\tau \odot m)}{\pi} \tag{58}$$

where the second line is an elementwise division. This is an algebraic identity at the population level. Accordingly, an estimator that replaces τ and m by consistent estimators $\widehat{\tau}, \widehat{m}$ and plugs in C converges to p. In particular, if C is known exactly and $\widehat{\tau} \odot \widehat{m}$ is (asymptotically) unbiased and consistent for $\tau \odot m$, then $(C^\top)^{-1}(\widehat{\tau} \odot \widehat{m})$ is (asymptotically) unbiased and consistent for $\pi \odot p$, and dividing elementwise by known π yields an unbiased consistent estimator of p.

Let $\widehat{n} := \widehat{\tau} \odot \widehat{m}$ be the empirical estimator of n. For a large number of identities, $\widehat{n}$ is approximately unbiased with some covariance matrix $\Sigma_{\widehat{n}} = \text{Cov}(\widehat{n})$. For the the (asymptotic) covariance of the corrected estimator we have:

$$\text{Cov}\left((C^\top)^{-1}\widehat{n}\right) = (C^\top)^{-1} \, \Sigma_{\widehat{n}} \, (C^\top)^{-T}. \tag{59}$$

Both sides are square matrix on Hilbert spaces thus, taking the operator norms, we have:

$$\left\| \text{Cov}\left((C^\top)^{-1}\widehat{n}\right)\right\|_{\text{op}} \leq \|(C^\top)^{-1}\|_{\text{op}}^2 \cdot \|\Sigma_{\widehat{n}}\|_{\text{op}}. \tag{60}$$

Thus the matrix norm $\|(C^\top)^{-1}\|_{\text{op}}^2$ controls the worst-case multiplicative inflation of covariance when inverting $C^\top$.If C tends toward identity, (i.e. the estimation of the attributes becomes perfect), then $\|(C^\top)^{-1}\|_{\text{op}} \to 1$ and there is thus no variance inflation. Conversely, if C has very small singular values (*i.e.* C is ill-conditioned), then $\|(C^\top)^{-1}\|_{\text{op}}$ is large and the variance of the corrected estimator can be dramatically larger than the variance of $\widehat{n}$.

□

C.3. Regularized Unbiased Estimator

C.3.1. Motivation: Numerical Instability at High Accuracy

When the confusion matrix C approaches the identity matrix due to high DAI accuracy, the estimator $\hat{p}^{(\text{corr})}$ from Proposition 2 exhibits numerical instability manifesting as high variance. This instability arises from three distinct mechanisms. First, the matrix inversion $(C^\top)^{-1}$ amplifies small estimation errors in C when the matrix is near-singular or when off-diagonal elements are estimated with noise on limited samples. Second, the element-wise division by the prior vector π magnifies errors disproportionately for minority demographic groups with small prior probabilities. Third, the empirical confusion matrix $\hat{C}$ itself suffers from estimation noise when computed on small validation folds. We address these instabilities through a three-stage regularization framework that introduces controlled bias-variance tradeoffs.

C.3.2. Three-Stage Regularized Estimator

We propose a hierarchical regularization scheme that stabilizes the estimator through confusion matrix shrinkage, prior regularization, and James-Stein combination.

Stage 1: Confusion Matrix Shrinkage. For a regularization parameter $\lambda_C \in [0, 1]$, we shrink the empirical confusion matrix toward the identity matrix:

$$C_{\text{reg}} = (1 - \lambda_C)\hat{C} + \lambda_C I \tag{61}$$

This shrinkage reduces the impact of estimation noise in off-diagonal elements, which are particularly unreliable when DAI accuracy is high (since misclassification events are rare).

Stage 2: Prior Regularization. For a regularization parameter $\lambda_\pi \in [0, 1]$, we define the regularized prior as a convex combination of the true prior π and the empirical observed proportions $\hat{\tau}$:

$$\pi_{\text{reg}} = (1 - \lambda_\pi)\pi + \lambda_\pi \hat{\tau} \tag{62}$$

The base corrected estimator is then:

$$\hat{p}^{(\text{corr},\lambda_C,\lambda_\pi)} = \frac{(C_{\text{reg}}^\top)^{-1}(\hat{\tau} \odot \hat{m})}{\pi_{\text{reg}}} \tag{63}$$

where the division is performed element-wise across all K demographic groups.

Stage 3: James-Stein Shrinkage. To further stabilize the estimator, we apply James-Stein shrinkage that combines $\hat{p}^{(\text{corr},\lambda_C,\lambda_\pi)}$ with the biased but stable estimator $\hat{m}$. The James-Stein estimator is:

$$\hat{p}^{(\text{JS})} = \hat{p}^{(\text{corr},\lambda_C,\lambda_\pi)} + \boldsymbol{\lambda}_{\text{JS}} \odot (\hat{m} - \hat{p}^{(\text{corr},\lambda_C,\lambda_\pi)}) \tag{64}$$

where $\boldsymbol{\lambda}_{\text{JS}} = (\lambda_{\text{JS},1}, \ldots, \lambda_{\text{JS},K})^\top$ is a vector of group-specific shrinkage intensities computed as:

$$\lambda_{\text{JS},g} = \max\left(0, \min\left(1, 1 - \frac{(K-2)\hat{\sigma}_g^2}{\sum_{k=1}^K (\hat{p}_k^{(\text{corr},\lambda_C,\lambda_\pi)} - \hat{m}_k)^2}\right)\right) \tag{65}$$

where $\hat{\sigma}_g^2$ is the estimated variance of $\hat{p}_g^{(\text{corr},\lambda_C,\lambda_\pi)}$ obtained via bootstrap resampling with B bootstrap iterations:

$$\hat{\sigma}_g^2 = \frac{1}{B-1}\sum_{b=1}^B \left(\hat{p}_{g,b}^{(\text{corr},\lambda_C,\lambda_\pi)} - \bar{p}_g^{(\text{corr},\lambda_C,\lambda_\pi)}\right)^2 \tag{66}$$

where $\bar{p}_g^{(\text{corr},\lambda_C,\lambda_\pi)} = \frac{1}{B}\sum_{b=1}^B \hat{p}_{g,b}^{(\text{corr},\lambda_C,\lambda_\pi)}$.

C.4. Binary Indicator and Fairness Metrics

There exist several metrics to estimate the fairness of face analysis systems (FAS), as presented in subsection B.2. For a group g, the False Match Rate (FMR) is "the rate at which a biometric process mismatches biometric signals from two distinct individuals as coming from the same individual" [52]. It is usually used for face verification, that is from a pair of face images, although some works consider this metrics for face recognition. For a pair of identities, it is defined as:

$$\text{FMR}_g = \Pr(\text{FAS accepts} \mid \text{pair is impostor}, G = g). \tag{67}$$

Common fairness metrics aims at reflecting the Demographic Parity, which ensures that the outcome of the FAS is the same across all groups defined by a sensitive attribute. Hence, the fairness can be estimated with the Demographic Parity Difference (DPD) and Demographic Parity Ratio (DPR) defined as

$$\text{DPD}_{\text{FMR}} = \max_{g,g'} |\text{FMR}_g - \text{FMR}_{g'}| \forall g, g' \in G \tag{68}$$

$$\text{DPR}_{\text{FMR}} = \frac{\min_g FMR_g}{\max_g FMR_g} \tag{69}$$

A difference close to 0 or a ratio close to 1 indicate a decision system fair with respect to the sensitive attribute across groups [8].

Another common fairness metric is the Degree of Bias (DoB), defined as the accuracy across different subgroups, which is higher when the performance varies a lot w.r.t each subgroup. Hence it can be expressed from the FMR as:

$$\text{DoB} = \sqrt{\frac{1}{K}\sum_{g=1}^K \left(1 - \frac{\text{FMR}_g}{\overline{\text{FMR}}}\right)^2}, \tag{70}$$

where $\overline{\text{FMR}}$ is the average FMR over groups.

The generic performance indicator $Y_i \in \{0, 1\}$ we consider in our paper can be specializer to express these metrics. If one evaluates the impostor trials such that $Y_i = \mathbb{1}\{\text{FRS accepts impostor trial}\}$, we have $\Pr(Y = 1 \mid G = g) = \text{FMR}_g$. If one rather evaluates the genuine trials then $Y_i = 1\{\text{FRS rejects genuine trial}\}$, thus we have the False Non-Match Rate (FNMR)[5]: $\Pr(Y = 1 \mid G = g) = \text{FNMR}_g$. Thus the group-specific expectation $p_g := \Pr(Y = 1 \mid G = g)$ is exactly the per-group error rate. The fairness metrics can thus be expressed accordint to the vector $p = (p_1, \ldots, p_K)$, *e.g.*:

$$\text{DPD}_{\text{FMR}} = \max_{g,g'} |p_g - p_{g'}| \tag{71}$$

$$\text{DoB} = \sqrt{\frac{1}{K}\sum_{g=1}^K \left(1 - \frac{p_g}{\overline{p}}\right)^2} \tag{72}$$

C.5. Fairness Benchmarking via Noise-Induced Degradation

To systematically evaluate the relationship between classifier performance and fairness metrics across different demo-

[5] i.e 1 - True Match Rate (TMR)

graphic annotation systems as well as theoretical considerations discussed in subsection 3.1, we propose a controlled degradation framework based on diffusion-based noise injection to simulate models with different accuracies.

C.5.1. Methodology

Let $\mathbf{y}_{\text{gt}} \in \{0,1\}^C$ denote the ground truth labels encoded as one-hot vectors, where C is the number of classes. We simulate classifier predictions with varying levels of accuracy by applying a diffusion process parameterized by $\alpha_t \in [0,1]$:

$$\mathbf{y}_t = \sqrt{\alpha_t} \cdot \mathbf{y}_{\text{gt}} + \sqrt{1-\alpha_t} \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

where α_t controls the signal-to-noise ratio . At $\alpha_t = 0$, predictions are pure Gaussian noise, while $\alpha_t = 1$ recovers the ground truth exactly. We extract the associated noisy label by taking the argmax of the considered noisy one-hot vector.

For each value of α_t, we compute classification accuracy alongside the corresponding estimated $\hat{m}$ (c.f. 2) as well as multiple fairness metrics, including demographic parity difference, demographic parity ratio, equalized odds difference, equalized odds ratio, and degree of bias. By varying α_t from 0 to 1, we generate continuous trajectories that characterize the theoretical relationship between DAI accuracy and, FAS fairness, FAS conditional success rate ($\hat{m}$ c.f. Equation 2), as well as $p^{(\text{corr})}$ (Proposition 2) under this degradation model. These trajectories are visualized with **DAI accuracy** on the **x-axis** and the considered **FAS-related metrics** on the **y-axis**, providing a direct DAI accuracy-FAS characterization landscape. Uncertainty bounds on trajectories can moreover be estimated through multiple noise realizations. Each landscape is drawn regarding 4 configurations obtained by considering as test set either RFW or BFW, and as FAS either an IR50 network trained on CASIA or a 1 million pictures subset drawn from GLINT.

Using the FAS trained on CASIA We also experimentally validate Proposition 1 by computing, as proxy of $\text{Bias}_g := m_g - p_g$, the biases $|\hat{p}_g - \hat{m}_g|$ as well as their corresponding upper bounds.

C.5.2. Annotator Evaluation

On top of the DAI accuracy-FAS characterization landscape described above, we evaluate four demographic DAI pipelines: FairFace, DeepFace, FaceXFormer, and our proposed annotator REDI-Face(IR50). For each annotator, we measure their classification accuracy and the considered metrics w.r. to the considered test dataset (RFW or BFW) and FAS model. These empirical measurements are then plotted directly on the DAI accuracy-FAS characterization landscape as individual points, with vertical lines indicating each annotator's accuracy level.

This approach enables direct visual comparison of real-world annotator performance against the simulated degradation curves. Annotators whose fairness metrics align with the theoretical curves exhibit fairness characteristics consistent with random noise degradation at their accuracy level, while deviations reveal systematic biases or properties independent of classification performance.

C.5.3. Results Analysis

Fas Fairness Metrics estimation : Figure 10 Figure 9, Figure 12, Figure 15, and Figure 17 show two interesting aspects:

- When baseline points lie on the degradation curve, it means they exhibit behaviors similar to the simulated noise models, i.e., their error might be modeled by Gaussian noise. Having an error that can be modeled by Gaussian noise is a crucial property, since it implies that the annotator satisfies 1.
- Some metrics are far more sensitive to DAI accuracy than others

We can thus notice that most baseline points are close to the simulated curves. Interestingly, behaviors become chaotic for the EOR estimation of DeepFace, and the DPD estimation of FaceXFormer. Our model shows a behavior that fits remarkably well with the Gaussian noise modeling.

Finally, it is clear that the most sensitive metric is the EOR ratio, which should therefore probably not be used for annotators with low accuracy.

Similarly as Figure 3, we report in Figure 19, Figure 20, and Figure 21 the absolute error between the fairness metrics computed with the baselines annotation and with our pipeline, REDI-Face. Overall, our approach exhibits far better Fairness metrics estimation than the other baselines.

Proposition 2 Analysis : Figure 11, Figure 14, Figure 16, and Figure 18, shows the L2 distance between $\hat{p}$ and $\hat{m}$ as well as the distance between $\hat{p}$ and $p^{(\text{corr})}$. It is important to realize that the computing $p^{(\text{corr})}$ requires to compute the confusion matrix C using true annotations. It can be done on a subset, but since we verify the theoretical part, the figures are computed on the entire test set.

It is interesting to note that, while annotators' behavior aligns with the modeling curve for fairness metrics, for the estimation $\hat{m}$, they no longer do, contrasting with our approach, REDI-Face.

The curves show that the unbiased estimator exhibits stable behavior with accuracy above 0.7 and that $p^{(\text{corr})}$ still works remarkably well on real annotators, despite not fitting the Gaussian model.

It thus appears that, providing the confusion matrix can be accurately computed (i.e., access to the true labels is possible), our unbiased estimator $p^{(\text{corr})}$ should be used instead of the biased estimator m. If not, using the most accurate

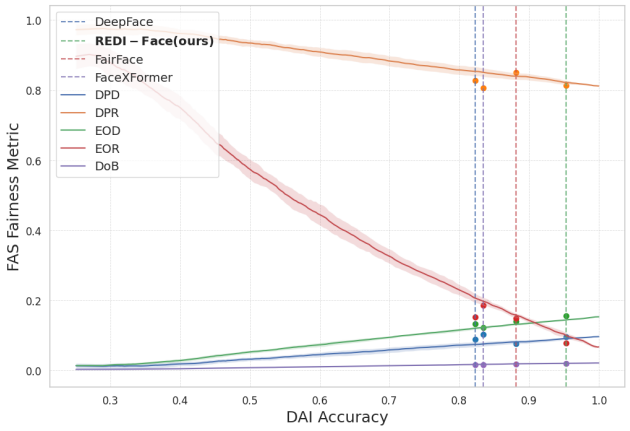


Figure 9. **FAS Fairness vs DAI Accuracy:**Evolution on RFW of the Fairness metrics of a FAS trained on CASIA w.r. to DAI Accuracy. The points correspond to the results on FairFace, FaceXFormer, REDI-Face, and DeepFace, identifiable through their abscissa (accuracy) value

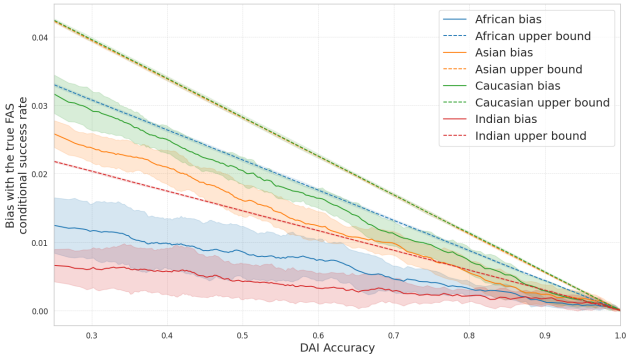


Figure 10. Experimental computation for a FAS trained on CASIA of the bias with p_g on RFW for each group and their corresponding evaluated theoretical upper bound (Equation 4)

annotator (i.e., REDI-Face here) is crucial.

Proposition 1 Validation : Figure 10 and Figure 13, show an evolution of biases and their corresponding upper bounds on RFW and BFW that are consistent with the theoretical results in Proposition 1. Few values exceeds the theoretical upper bounds, which is very likely linked to the approximation error, since we are not considering the theoretical values m and p but their estimates $\hat{m}$ and $\hat{p}$. Interestingly enough, the estimation of the upper bounds appear more stable than the estimation of the Bias. The results experimentally confirm Proposition 1, highlighting the important role of accurate DAI at estimating m for a given FAS model.

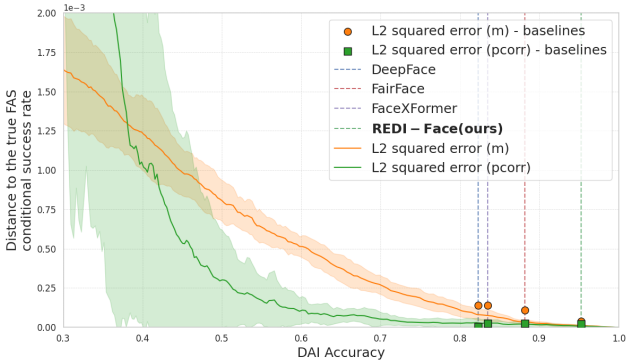


Figure 11. Distance to the FAS (trained on CASIA) true p on RFW when considering pcorr (Proposition 2.) or m, the uncertainty halo is obtained by computing the standard deviation over multiple noises. The points correspond to the results on FairFace, DeepFace, and REDI-Face(ours), identifiable through their abscissa (accuracy) value

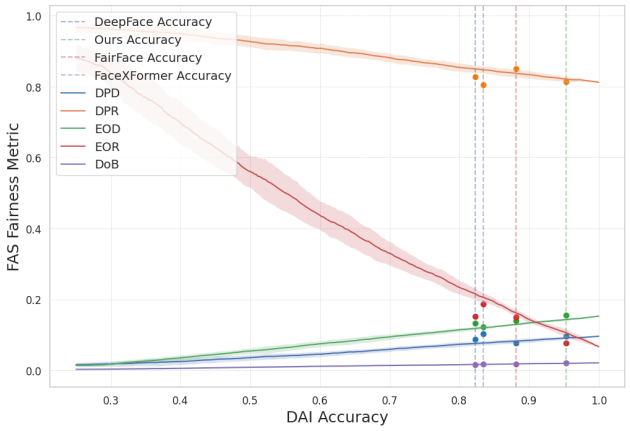


Figure 12. **FAS Fairness vs DAI Accuracy:**Evolution on BFW of the Fairness metrics of a FAS trained on CASIA w.r. to DAI Accuracy. The points correspond to the results on FairFace, FaceXFormer, REDI-Face, and DeepFace, identifiable through their abscissa (accuracy) value

D. Marginal Effects Analysis

To rigorously quantify the impact of training configuration choices on DAI accuracy performance, we conduct marginal effects analysis using ordinary least squares (OLS) regression. This approach allows us to estimate the average causal effect of each variable while controlling for confounders, providing a more robust understanding than simple bivariate comparisons.

D.1. Methodology

D.1.1. Model Specification

We specify two related regression models to analyze different aspects of our experimental results:

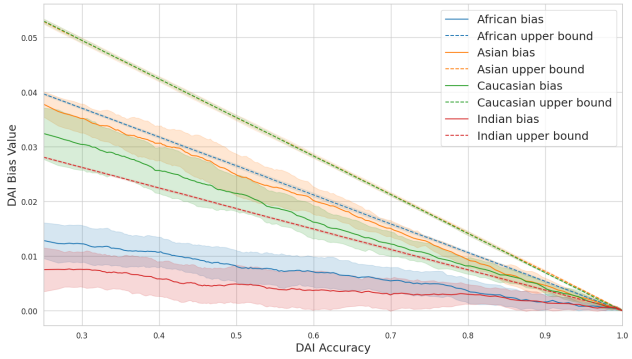


Figure 13. Experimental computation for a FAS trained on CASIA of the bias with p_g on BFW for each group and their corresponding evaluated theoretical upper bound (Equation 4)

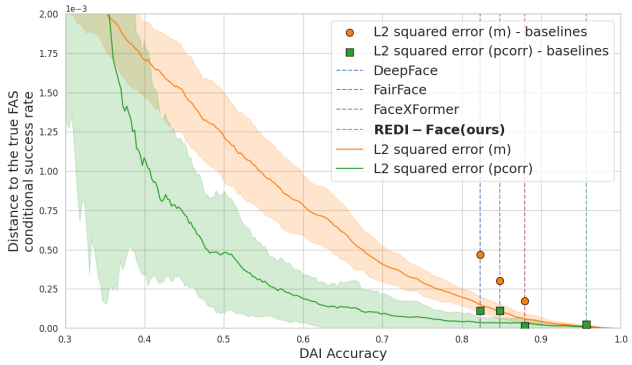


Figure 14. Distance to the FAS (trained on CASIA) true p on BFW when considering pcorr (Proposition 2.) or m, the uncertainty halo is obtained by computing the standard deviation over multiple noises. The points correspond to the results on FairFace, DeepFace, and REDI-Face(ours), identifiable through their abscissa (accuracy) value

Model 1: Best Configuration Analysis For the first analysis, examining the best performing configuration per architecture, we model the target DAI accuracy metric y_i (best RFW validation accuracy at optimal epoch) as:

$$y_i = \beta_0 + \beta_1 \log(\text{IDs}_i) + \beta_2 \log(\text{Samples}_i) + \beta_3 \text{FV}_i + \beta_4 \text{Unbal}_i + \epsilon_i$$

where:

- IDs_i is the number of identities in the training set
- Samples_i is the number of samples per identity
- FV_i is the face verification accuracy of the base model
- Unbal_i is a binary indicator for unbalanced training (vs. balanced baseline)
- $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ is the error term

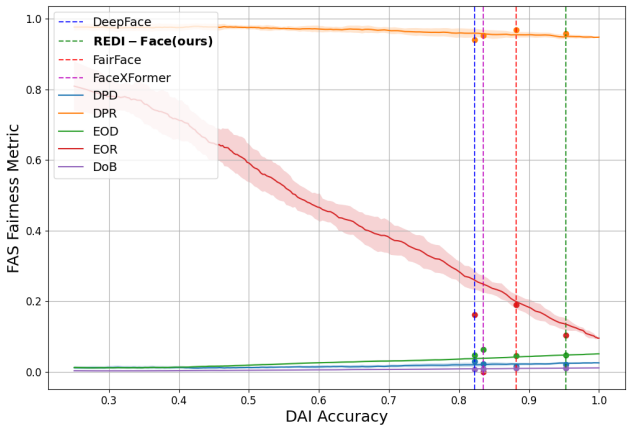


Figure 15. **FAS Fairness vs DAI Accuracy:**Evolution on RFW of the Fairness metrics of a FAS trained on GLINT w.r. to DAI Accuracy. The points correspond to the results on FairFace, FaceXFormer, REDI-Face, and DeepFace, identifiable through their abscissa (accuracy) value

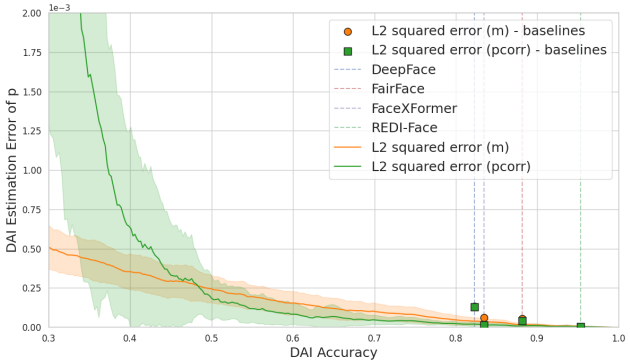


Figure 16. Distance to the FAS (trained on GLINT) true p on RFW when considering pcorr (Proposition 2.) or m, the uncertainty halo is obtained by computing the standard deviation over multiple noises. The points correspond to the results on FairFace, DeepFace, and REDI-Face(ours), identifiable through their abscissa (accuracy) value

Model 2: Per-Epoch Analysis For the second analysis examining performance across training epochs for specific evaluation sets, we extend the model to:

$$y_{iet} = \beta_0 + \beta_1 \log(\text{IDs}_i) + \beta_2 \log(\text{Samples}_i) + \beta_3 f(\text{Epoch}_t) + \beta_4 C_i + \beta_5 \text{Unbal}_i + \epsilon_{iet}$$

where y_{iet} is the accuracy for configuration i, evaluation set e, at epoch t, and C_i represents either:

- A continuous control variable (FV accuracy on a specific benchmark), or
- A set of architecture dummy variables: $\sum_{k=2}^{K} \gamma_k \mathbb{1}(\text{Arch}_i = k)$

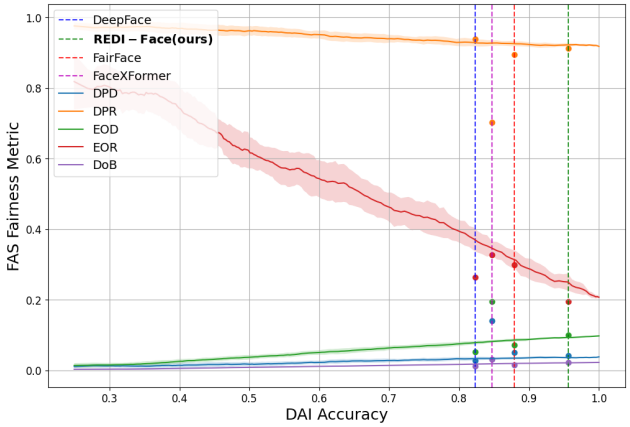


Figure 17. **FAS Fairness vs DAI Accuracy:**Evolution on BFW of the Fairness metrics of a FAS trained on GLINT w.r. to DAI Accuracy. The points correspond to the results on FairFace, FaceXFormer, REDI-Face, and DeepFace, identifiable through their abscissa (accuracy) value

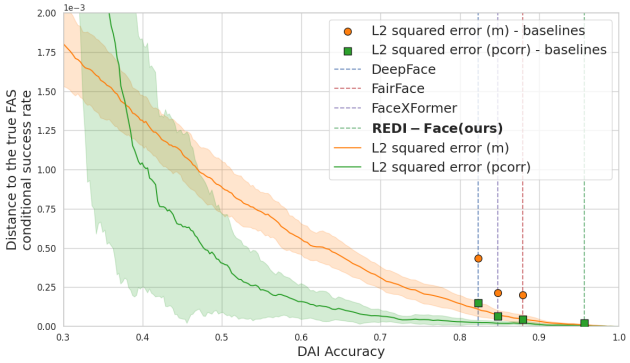


Figure 18. Distance to the FAS (trained on GLINT) true p on BFW when considering pcorr (Proposition 2.) or m, the uncertainty halo is obtained by computing the standard deviation over multiple noises. The points correspond to the results on FairFace, DeepFace, and REDI-Face(ours), identifiable through their abscissa (accuracy) value

The function $f(\cdot)$ applies logarithmic transformation if epochs span more than one order of magnitude, otherwise uses linear scaling.

D.1.2. Variable Transformations

To ensure proper model specification and interpretability, we apply the following transformations:

Log Transformations Variables spanning multiple orders of magnitude (number of training identities: 500–20,000; samples per identity: 1–40) are log-transformed. This addresses two concerns:

1. **Heteroskedasticity**: Raw values exhibit increasing variance with scale

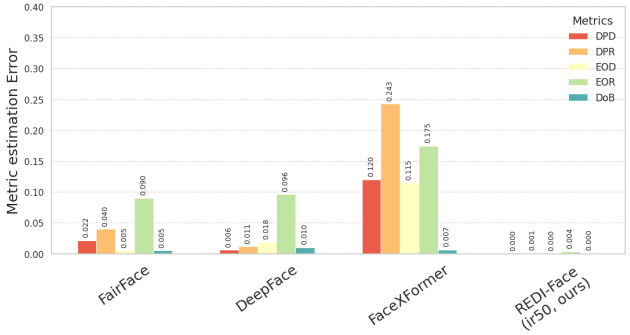


Figure 19. Absolute error between the fairness metrics on BFW and the ones estimated with annotators when considering a FAS trained on CASIA.

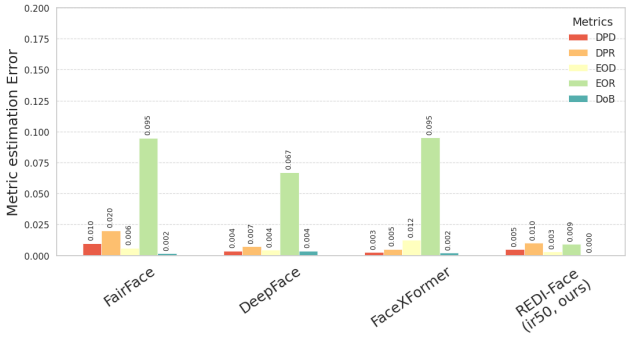


Figure 20. Absolute error between the fairness metrics on RFW and the ones estimated with annotators when considering a FAS trained on GLINT.

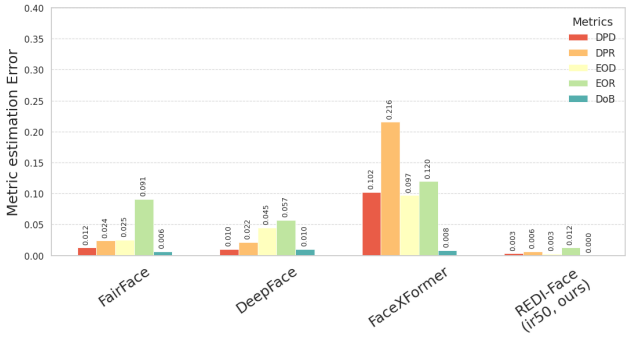


Figure 21. Absolute error between the fairness metrics on BFW and the ones estimated with annotators when considering a FAS trained on GLINT.

2. **Interpretability**: Coefficients represent *elasticities* (percentage changes) rather than absolute changes

For a log-transformed variable, the marginal effect of doubling is computed as:

$$\Delta y_{\text{double}} = \beta_{\log(x)} \times \log(2) \approx 0.693\beta_{\log(x)} \quad (73)$$

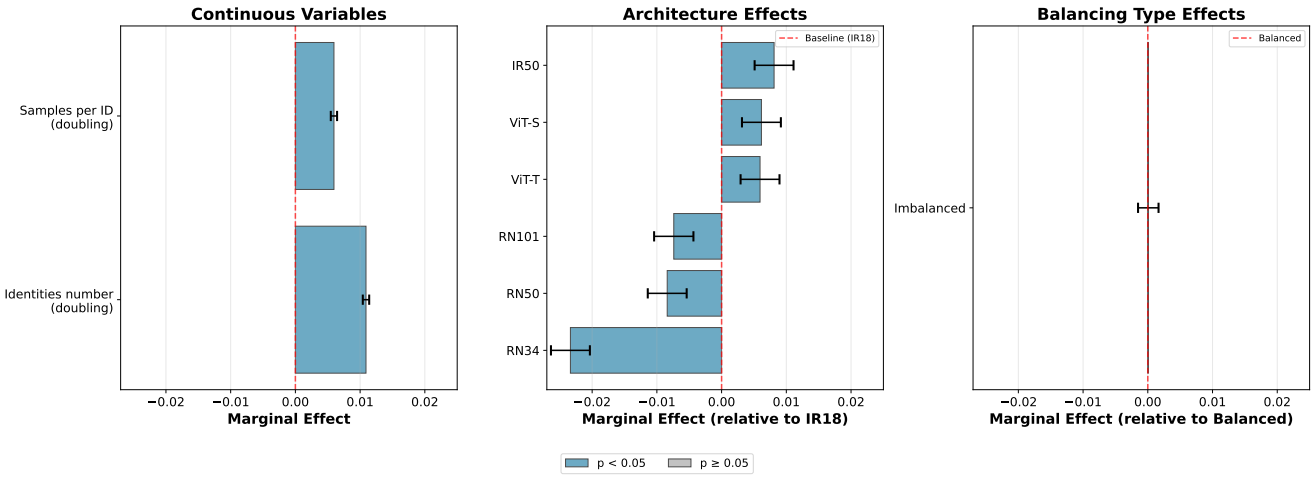


Figure 22. Marginal effects for OLS Model 2 on RFW

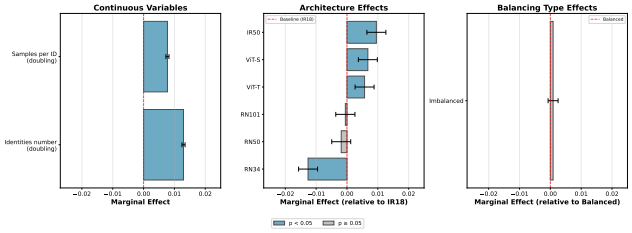


Figure 23. Marginal effects for OLS Model 2 on BFW

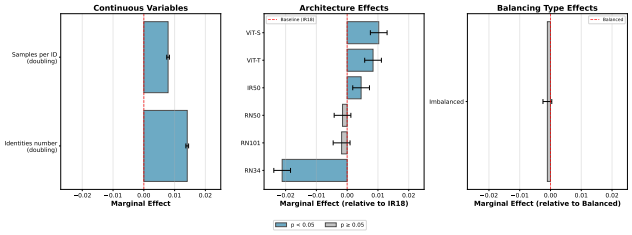


Figure 24. Marginal effects for OLS Model 2 on BUPT

Categorical Variables Categorical variables (architecture, balancing type) are dummy-coded with one category as the baseline. The coefficient γ_k represents the average difference in outcome between category k and the baseline, holding all other variables constant.

D.1.3. Why Control for Face Verification Accuracy?

A critical methodological consideration is whether to include architecture as a categorical variable or control for face verification (FV) accuracy. We observe strong collinearity between architecture and FV performance ($r > 0.85$), as different architectures inherently achieve different base recognition capabilities.

This creates two valid but distinct research questions:

- **Controlling for architecture**: "Which training configurations work best for each architecture?"
- **Controlling for FV accuracy**: "Given a certain base recognition quality, how do training choices affect DAI accuracy?"

For Model 1, we control for FV accuracy rather than architecture because:

1. Architecture and FV accuracy measure fundamentally the same construct (model capability)
2. Including both would introduce multicollinearity, inflating standard errors
3. FV accuracy provides a continuous, more granular measure of model quality
4. The coefficient on FV accuracy has a natural interpretation: DAI accuracy as a function of base recognition capability

For Model 2, we allow flexibility to examine either control strategy depending on the research question.

D.2. Results

D.2.1. Model 1: Optimal Configuration Analysis

Figure 25, Figure 26, and Figure 27 present the marginal effects from Model 1. The regression achieves $R^2 = 0.856$ ($R^2_{\text{adj}} = 0.853$), indicating excellent model fit.

Training Set Size Doubling the number of training identities yields a **+1.21 percentage point** increase in DAI accuracy ($p < 0.001$). This substantial effect demonstrates that DAI performance scales predictably with dataset diversity, consistent with prior work showing that representation matters for bias mitigation [7].

Samples per Identity Doubling samples per identity increases accuracy by **+0.64 percentage points** ($p < 0.001$).

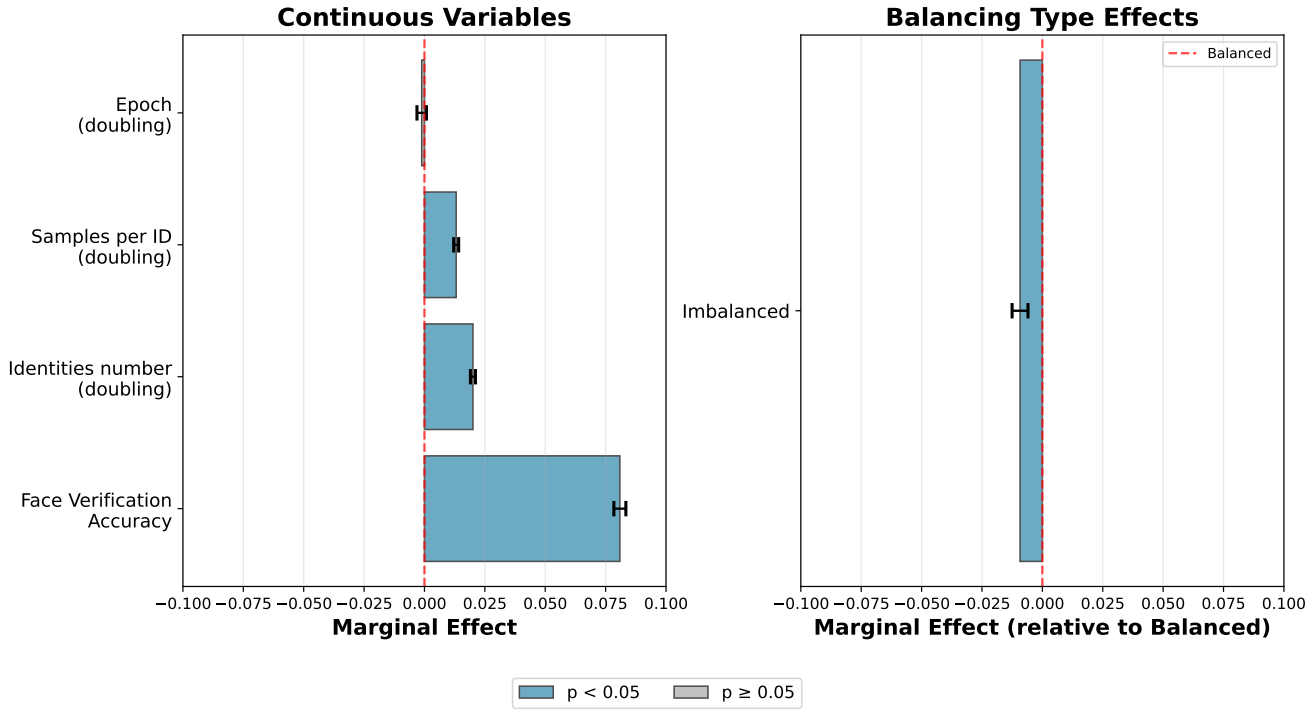


Figure 25. Marginal effects for OLS Model 1 on RFW

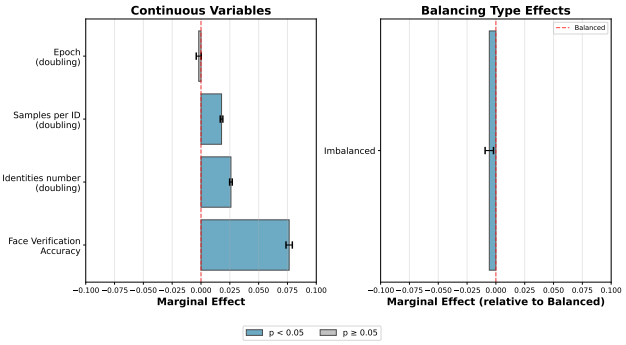


Figure 26. Marginal effects for OLS Model 1 on BFW

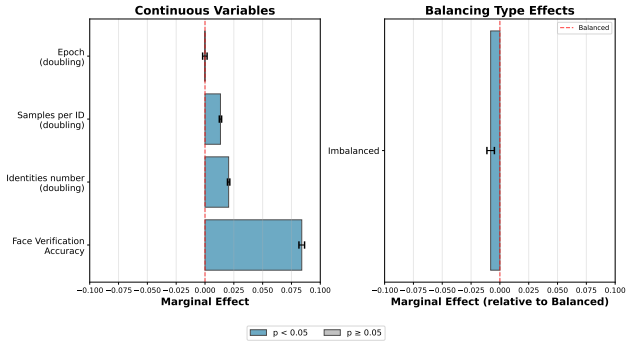


Figure 27. Marginal effects for OLS Model 1 on BUPT

While significant, this effect is roughly half that of adding identities, suggesting that *diversity* (more identities) is more valuable than *depth* (more samples per identity) for DAI accuracy.

Base Model Quality A 10-point increase in base FV accuracy corresponds to a **+5.43 percentage point** improvement in DAI performance ($p < 0.001$). This is the strongest predictor in the model, with an elasticity of approximately 0.54, indicating that DAI performance is highly sensitive to base recognition quality.

This finding has important implications: improving the underlying face verification model (through better architectures, pretraining, or optimization) yields larger DAI accuracy gains than dataset manipulation alone.

Balancing Strategy Unbalanced training shows a small but significant **positive** effect (+0.26 percentage points, $p = 0.003$) compared to balanced training. This counterintuitive result suggests that naive balancing may not always improve DAI accuracy.

D.2.2. Model 2: Per-Epoch Analysis

With the Model 2 Analysis, we confirm the resulting performances for the different backbones obtained in Table 1. We also observe the same behavior for the impact of the num-

ber of identities and the number of samples per identity as with the Model 1.

D.3. Marginal effects Conclusion

Marginal effects analysis reveals that the backbone choice (FV accuracy/architecture) is the strongest predictor of annotator performance, followed by training set diversity (number of identities) and depth (samples per identity). Surprisingly, when controling on the face verification accuracy, unbalanced training slightly outperforms naive balancing, thus showing a difference between ”In Average behaviors” and ”Maximum reachable Performances”.